

Material-Dependent Evolution of Electromagnetic Shower Geometry

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Why do we want to go to CERN?

We are united by one question: does electromagnetic shower geometry change with the material? After in-depth theory and GEANT4 simulations with 3 metals, we seek real beamline data. CERN provides a high-energy electron beam and precision detectors that capture shower complexity beyond simulations. This experiment combines detector physics, computational analysis, and calorimetry, and CERN is the only place where we, as high school students, can turn theory into real science. For us, it's more than a competition—it's seeing our work come alive.

Project Objectives

Accurate modelling of electromagnetic calorimetry is key in high-energy physics, but whether shower geometry is universal across absorber materials remains untested. This study investigates if EM shower morphology, via fractal dimension and graph-based clustering, is material-independent for aluminium, copper, and lead using GEANT4 simulations and CERN beamline data.

Introduction

Electromagnetic shower development in dense absorbers is well understood theoretically, but whether material properties produce distinct geometries remains untested. Aluminum ($Z=13$), copper ($Z=29$), and lead ($Z=82$) have radiation lengths from 8.99 to 0.56 cm and Molière radii from 4.45 to 1.61 cm, showing significant geometric differences. High-energy electrons initiate showers via bremsstrahlung and pair production, multiplying secondaries until they fall below the critical energy (E_c), then attenuating. This study investigates how absorber choice affects shower geometry using fractal dimension analysis and graph-based clustering.

Technical Parameters & Shower Physics

Our setup is optimized for 1–6 GeV electron beams, ensuring precise measurement of electromagnetic shower development.

Radiation Length (X_0) is calculated from the electron density of the material and is proportional to it for most materials:

$$X_0 \propto \frac{A}{Z^2 \ln \left(\frac{183}{Z^{1/3}} \right)}$$

Composite Materials (Lead Glass)

Since lead glass is a mixture. We use effective atomic numbers (Z_{eff}) and mass number (A_{eff}), calculated via weight fraction (f_i):

- **Effective Atomic Number:**

$$Z_{\text{eff}} = \frac{\sum f_i Z_i}{\sum f_i}$$

- **Effective Mass Number:**

$$A_{\text{eff}} = \frac{\sum f_i A_i}{\sum f_i}$$

Given Values:

- $Z_{\text{eff}} = 50$
- $A_{\text{eff}} = 14$

Empirical Dahl formula for calculate the absolute radiation length g/cm^2 ,

$$X_0 \text{ (g/cm}^3\text{)} = \frac{716.4 A_{\text{eff}}}{Z_{\text{eff}}(Z_{\text{eff}} + 1) \cdot \ln \left(\frac{287}{\sqrt{Z_{\text{eff}}}} \right)}$$

X_0 is the “step size” of the shower, the average distance an electron travels before interacting with lead glass to produce new particles. The 37 cm calorimeter equals ~14.8 radiation lengths, sufficient for measuring electron energy.

Parameters	Values
$X_0 \text{ (g/cm}^2\text{)}$	8.66 g/cm²
$X_0\text{(cm)}$	8.66/3.86=2.24cm
Lead Glass Density (<i>P</i>)	$\approx 3.86\text{g/cm}^2$
Radiation Length of lead glass X_0	$8.66\text{g/cm}^3 \approx 2.24\text{cm}$

Critical Energy:

$$E_c = \frac{550\text{MeV}}{Z_{\text{eff}}} = \frac{550\text{MeV}}{50} = 11\text{MeV}$$

11. MeV is the energy at which particle multiplication stops, marking the lower limit of the multiplication phase. When the average energy of secondary particles drops below E_c ionization dominates, particle count rapidly decreases, and the shower dies out.

Moliere Radius:

$$R_m \approx \frac{E_s}{E_c} X_0$$

Where E_s is the scattering energy $E_s \approx 21\text{MeV}$.

$$R_m \approx \frac{21\text{MeV}}{11\text{MeV}} \times 2.24\text{cm} \approx 4.27\text{cm}$$

Material	Atomic Number (Z)	Radiation Length (X_0)	Critical Energy (E_c)	Moliere Radius (R_m)
Aluminium	13	~8.99 cm	42.84 MeV	4.45 cm
Copper	29	~1.47 cm	20.17 MeV	1.54 cm
Lead	82	~0.56 cm	7.33 MeV	1.61 cm

Shower Maximum:

For our peak beam energy of 6 GeV, the depth of maximum particle multiplication is predicted as

We use a logarithmic formula

$$X_{\max} = X_0 \cdot \ln\left(\frac{E_a}{E_c}\right)$$

Parameters:

1. Beam Properties:

Energy Range	1 GeV to 6 GeV
Particle type	Pure Electrons
Beam Spot Size	Round Cross Section with a diameter of 2cm at the focal point.
Intensity/Rate	We will target a rate of particles within the CERN's Data acquisition (DAQ) limits

2. Tracking Parameters:

To determine the fractal dimension & spatial entropy we need precise energy coordinates.

Quantity	Four DWC will be used
Sensitive Area	10 cm × 10 cm
Spatial Resolution	200 μm – 300 μm

3. Calorimeter parameters (lead glass ECAL)

Material	High-density lead glass
Block Dimension	10 cm It will be placed by matching the DWC area and 37 cm length.
Total Radiation Length(X_0)	Approximately 16.5 consuming
Energy Measurement	The total energy of electrons/photons via Cherenkov light production.

4. Theoretical Prediction of Shower Maximum (Xmax)

The shower maximum is the depth where the number of secondary particles and spatial complexity peak. For a 4 GeV electron, Xmax is calculated as follows:

$$E_0 = 4000 \text{ MeV}$$

$$E_c = 11 \text{ MeV}$$

$$C = -0.5 \text{ (for electron)}$$

$$t_{\max} = \ln \left(\frac{E_0}{E_c} \right) + C$$

$$t_{\max} = \ln \left(\frac{4000}{11} \right) - 0.5 = 5.3 X_0$$

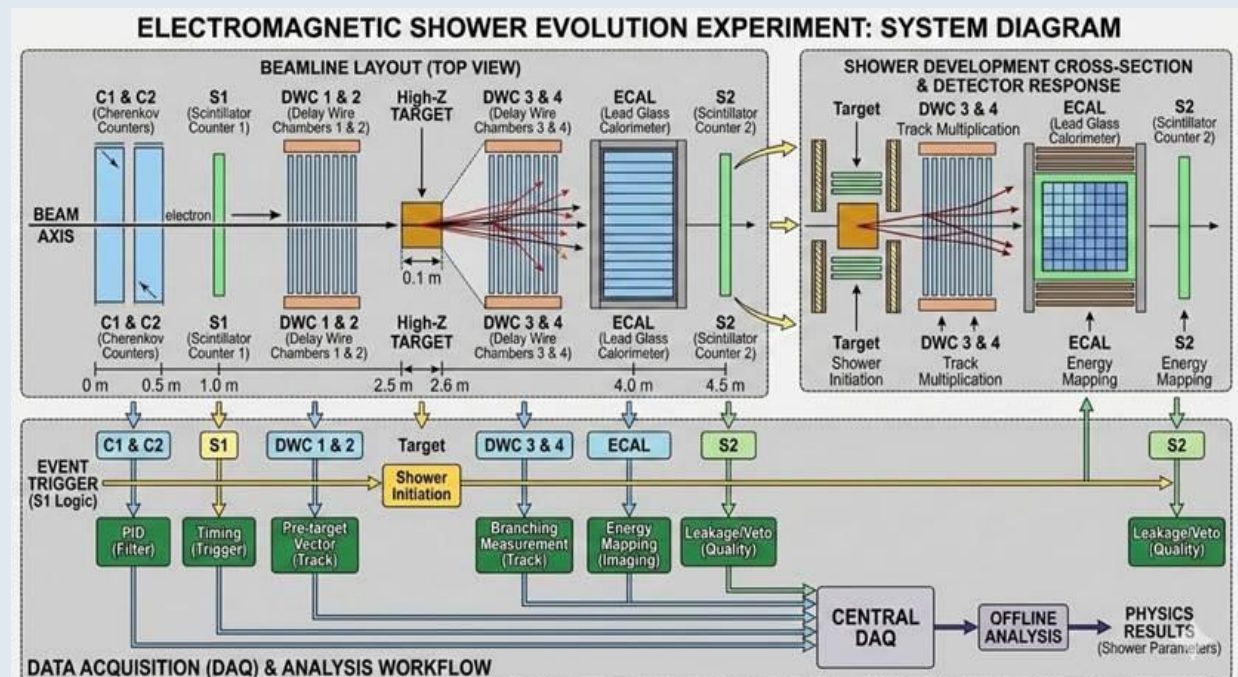
Converting to physical depth,

$$X_{\max} = 5.3 \times 2.24 = 12 \text{ cm}$$

Conclusion:

Since our detector is 37Cm deep, The Shower maximum(12Cm) Occurs well within the detectors which confirms that our detectors can fully generate the most complex part of the shower without any data loss.

Detectors layout



Zone-1 Upstream Tracking	Zone-2 The Interaction zone	Zone-3 Downstream Analysis
In this zone, particles are identified with their initial velocity (X, Y position, & angle) and are measured before it hits the target. Fixed threshold Cherenkov detectors (C₁, C₂) Scintillators S₁ Delay wire chamber (DWC₁ & C₂)	The Initial zone of shower born Removable Target slides: A mechanical Helder containing our two primary experimental variables: 5mm Copper (Cu): To study Young low-density shower (~0.35X₀) 5mm lead (Pb): To study a Mature high-density shower (~0.89X₀) DWC₃ & C₄	This Zone provides the data of fractal analysis Lead Glass Calorimeter (ECAL): A 3 × 3 or 5 × 5 array of lead glass block. Scintillator S₂(Veto/Leakage):

For the detailed detectors information please check out this [repository](#)

Simulation description

In GEANT4, we simulated a 20×20 cm absorber at various depths, with a 20×20 cm detector placed 1.5 m behind.

Each simulation fired 100,000 electrons perpendicularly.

Three absorber materials were used: lead, copper, aluminium.

For each material, six runs varied initial electron energy from 1-6 GeV.

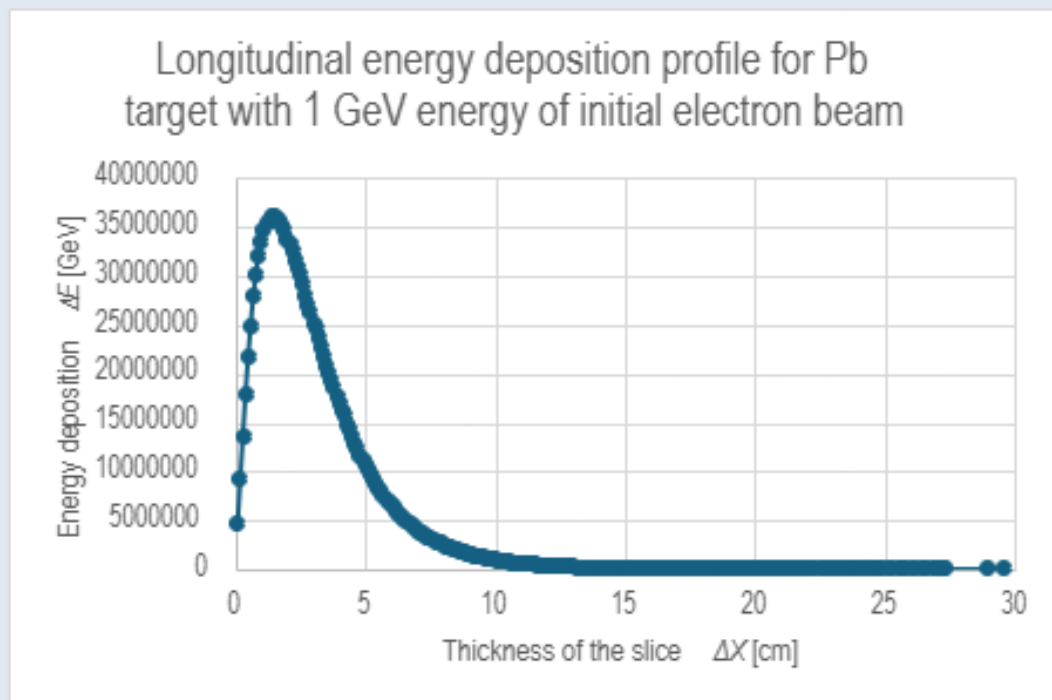
Multiple simulations tested target depths (ΔX) from 1 mm to 30 mm.

The detector, 1.5 m behind the target, had 10×10 cells (10×10 cm each), totalling 100×100 cm detection area.

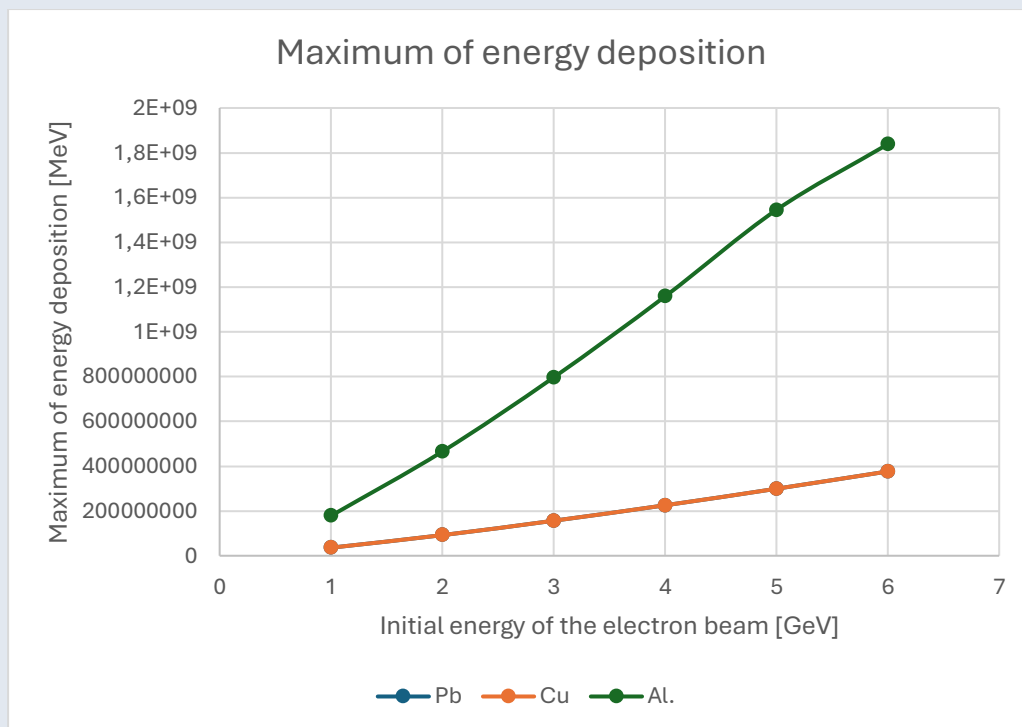
Data Analysis

Longitudinal energy deposition profile

We have prepared several longitudinal energy deposition profiles like the one below. All of them can be seen in appendix A.

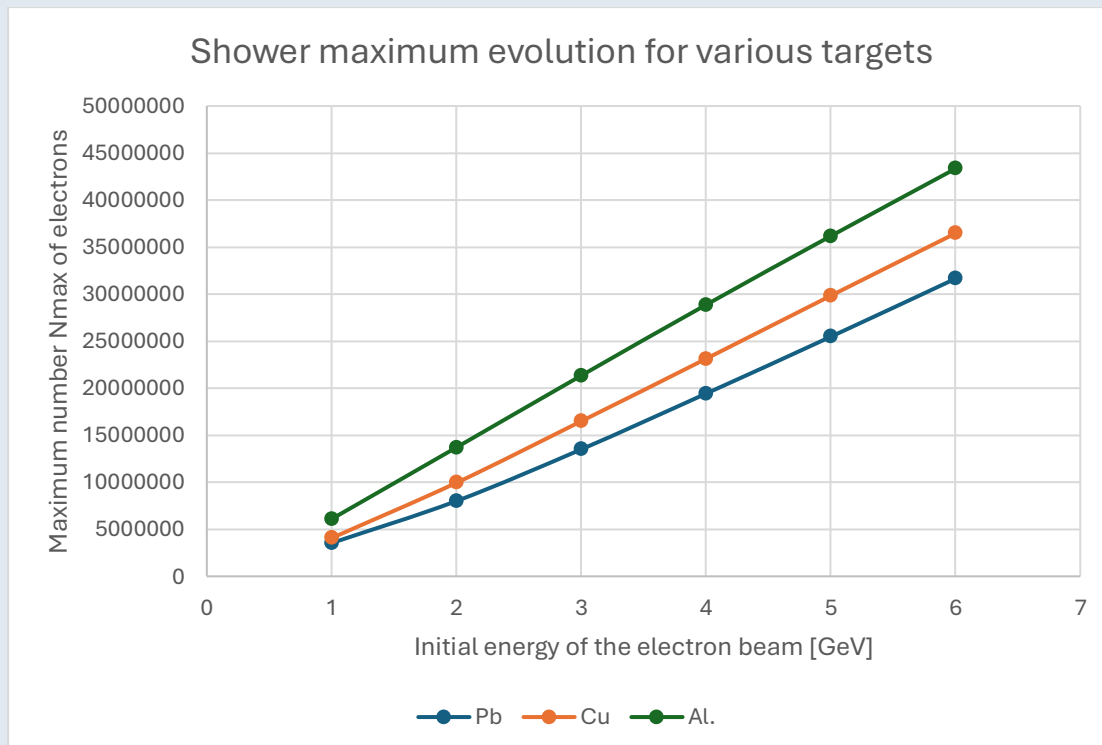


From each profile, we extracted the peak energy deposition and its depth ($X_{\max}$). Cu and Pb show similar peaks, while Al reaches significantly higher values. Energy deposition increases with beam energy for all materials.

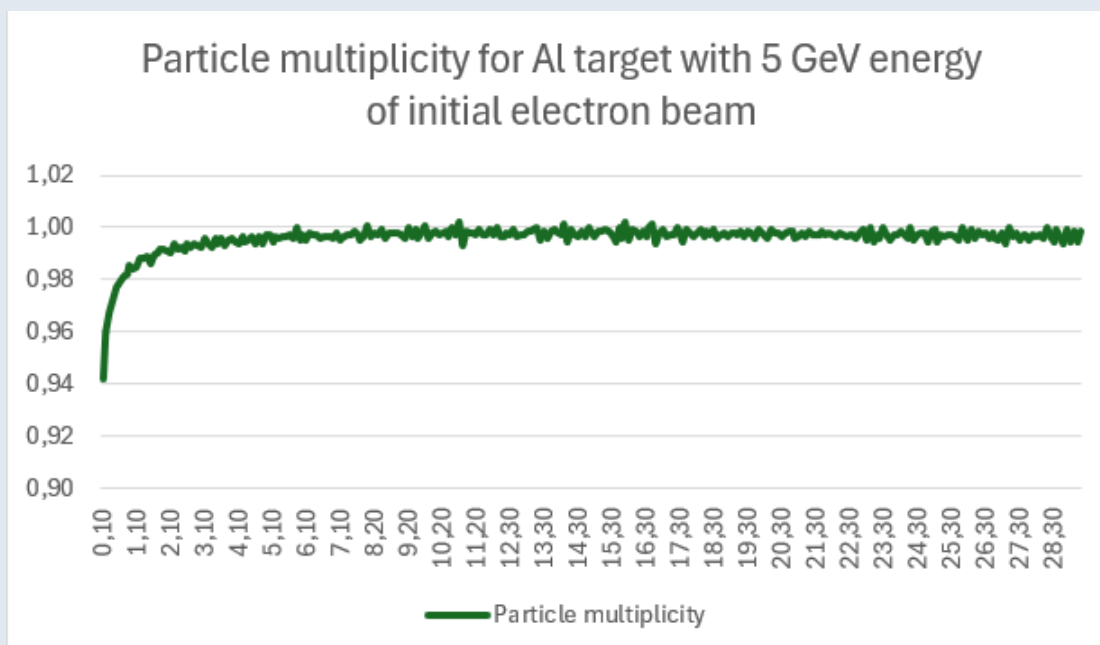


Shower maximum X_{max}

Maximum number of electrons in cascade increase with initial energy of the beam. The highest is for aluminium, then for copper and lead.



Particle multiplicity



The particle multiplicity graph indicates our detector captured only a small central fraction of cascade electrons. A new simulation should show multiplicity first above 1 (growth) then below 1 (decay), unlike our current graphs, which drop immediately and then stabilize.

Radial energy deposition profiles

Radial energy deposition profiles are presented in appendix B. The main trend is decreasing, although some fluctuations are also visible.

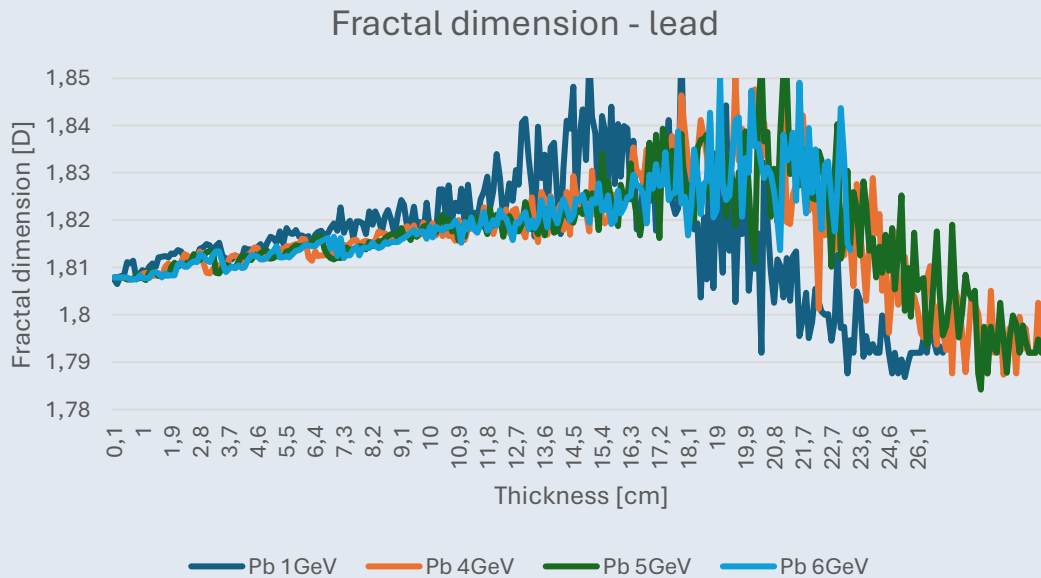
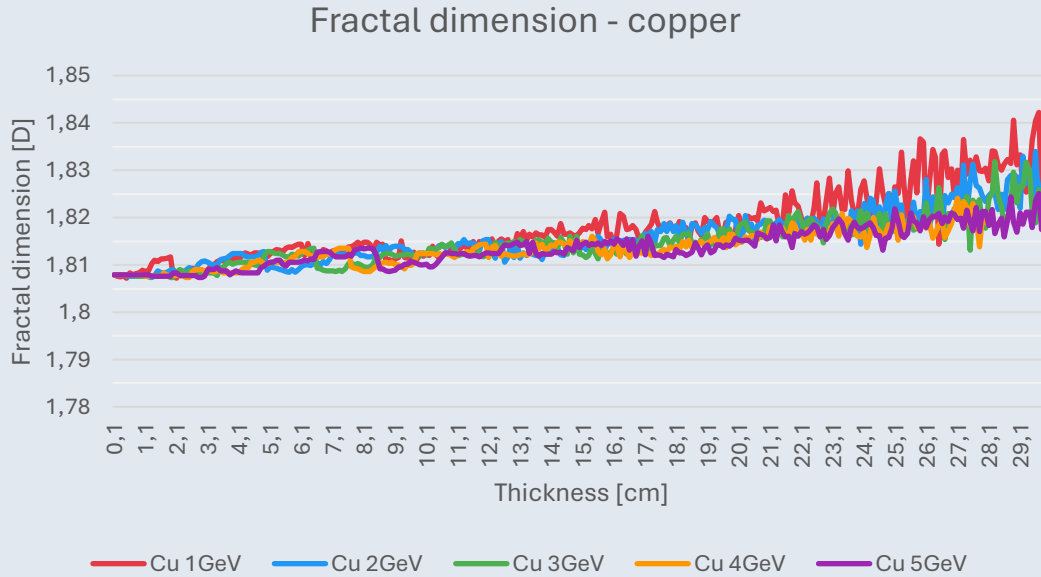
Entropy of the shower distribution

In the table below one can see that entropy of the shower increases with initial energy of the beam. Entropy is the highest for Aluminium, then for copper, lead.

	1GeV	2GeV	3GeV	4GeV	5GeV	6GeV
Cu	4,566	4,68	4,723	4,759	4,785	4,799
Pb	3,748	3,836	3,866	3,909	3,928	3,944
Al	5,634	5,66	5,665	5,666	5,663	5,649

Fractal dimension analysis

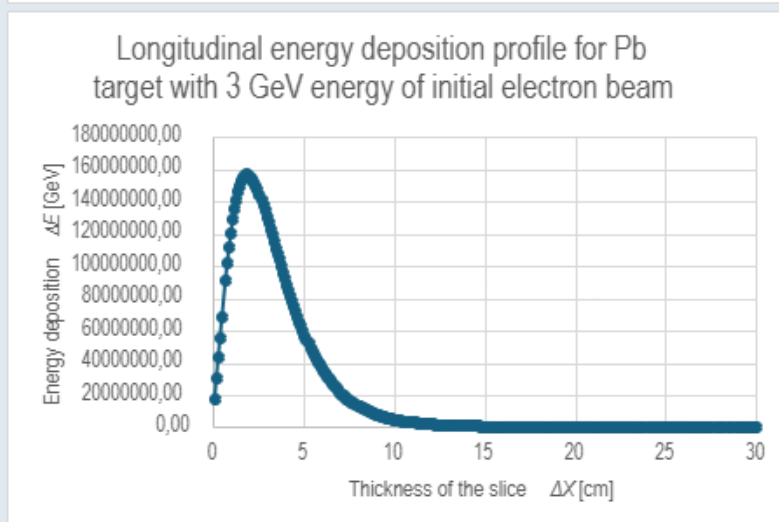
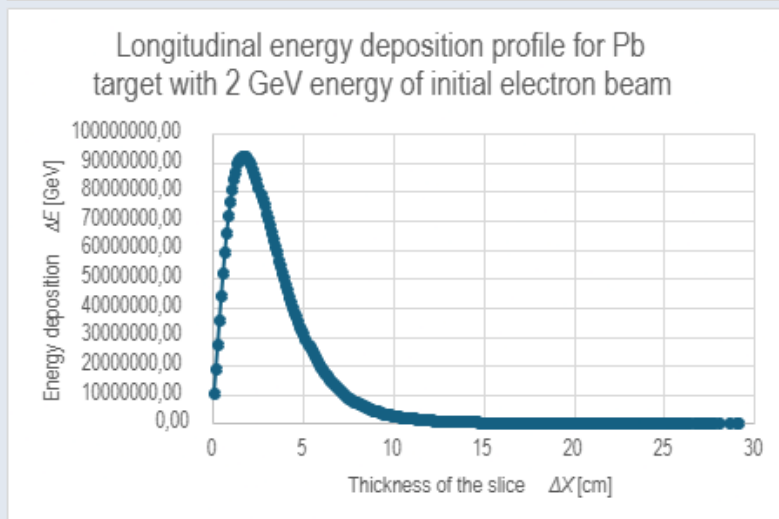
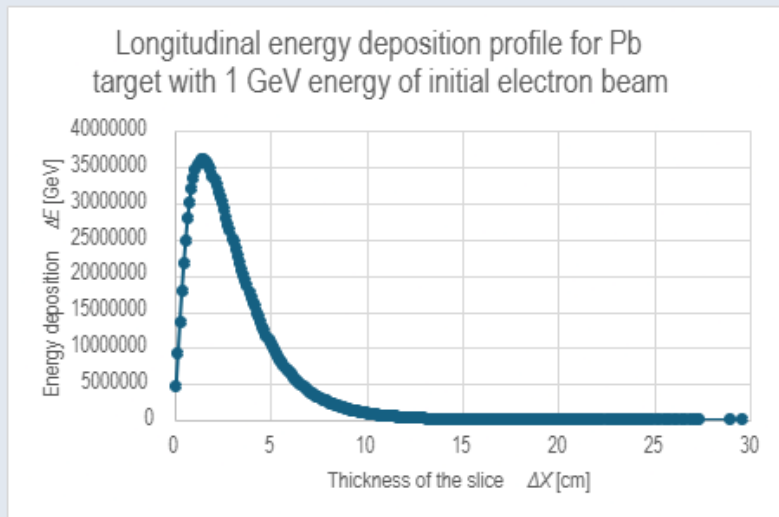
The fractal dimension D of the detector hit distribution was computed via 3D box-counting. The hit map used X and Y as detector coordinates and Z as normalized hit count $[0,1]$. Boxes of size ϵ covered the space, counting $N(\epsilon)$ needed at each scale. D was the slope of $\log(N)$ vs $\log(1/\epsilon)$, with R^2 assessing fit quality.



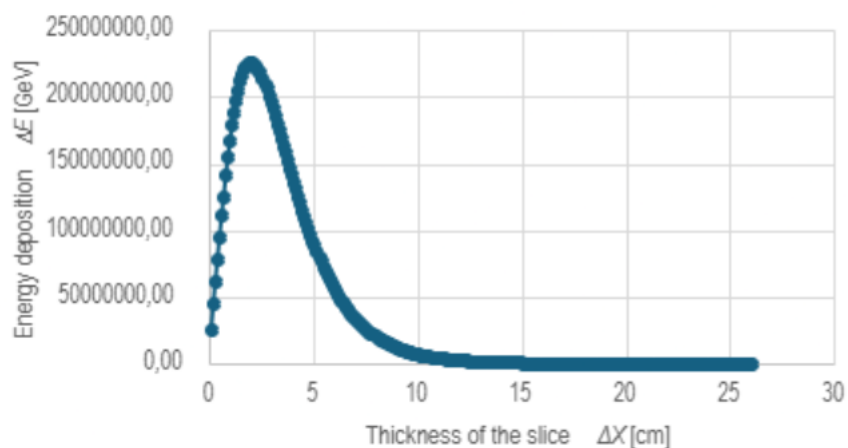
APPENDICES

APPENDIX A – Longitudinal Energy Deposition Profiles

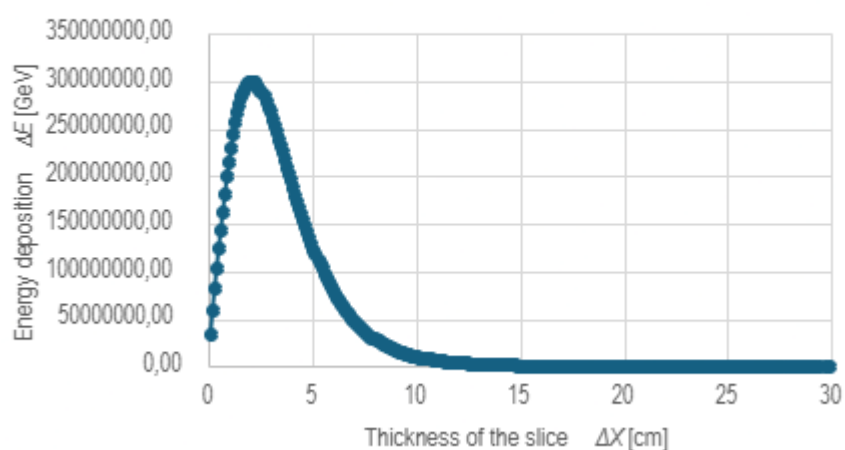
Pb:



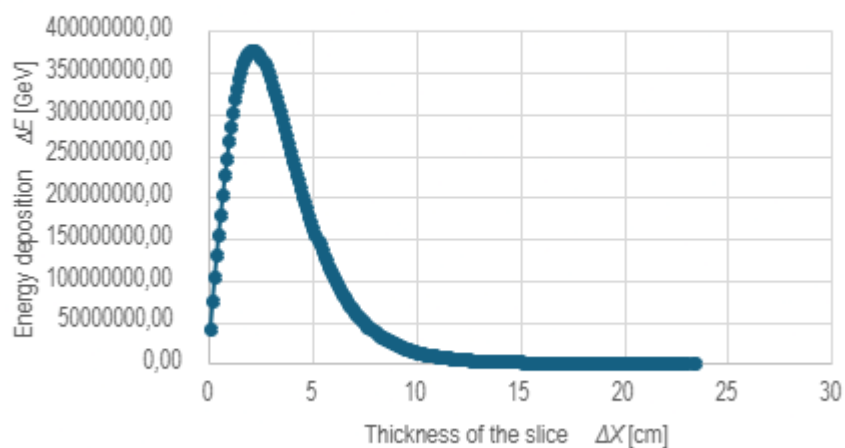
Longitudinal energy deposition profile for Pb target with 4 GeV energy of initial electron beam



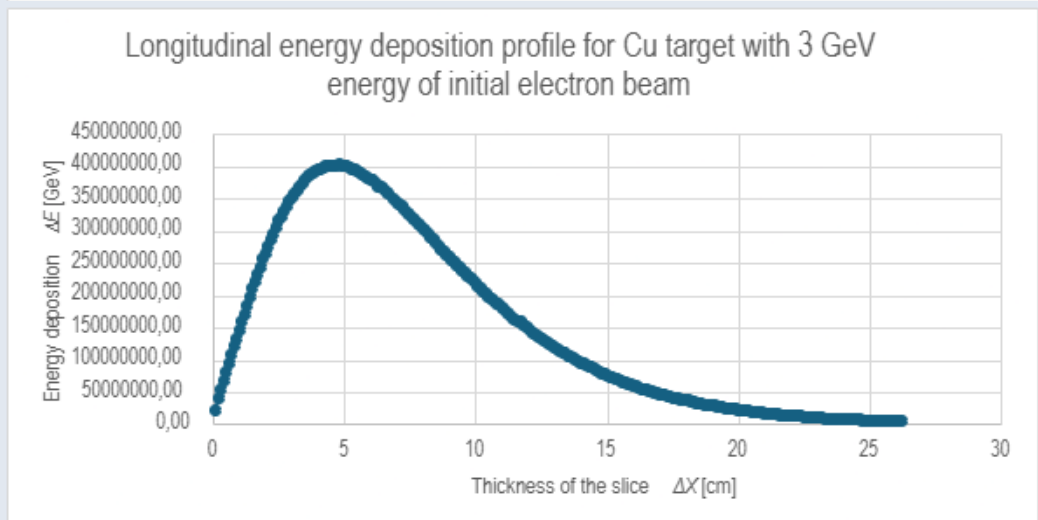
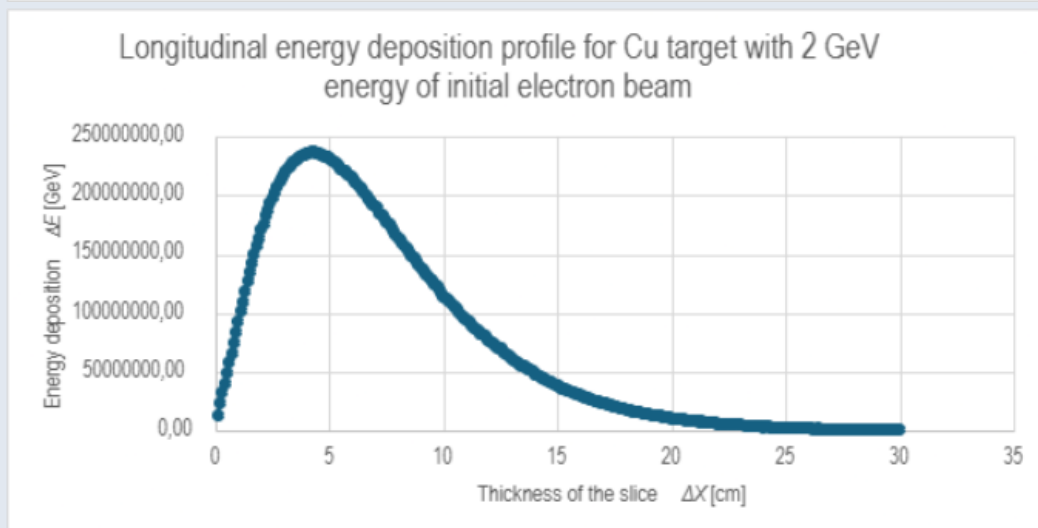
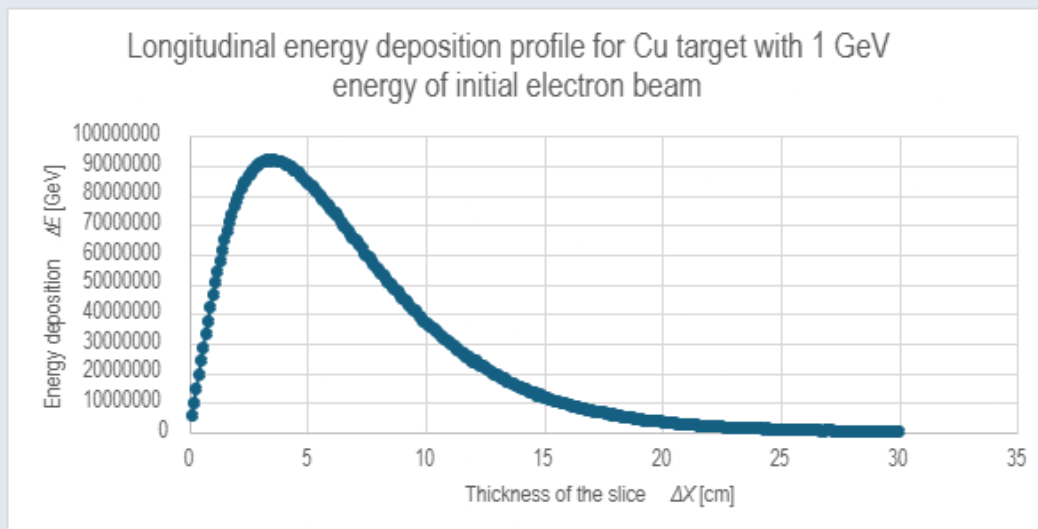
Longitudinal energy deposition profile for Pb target with 5 GeV energy of initial electron beam



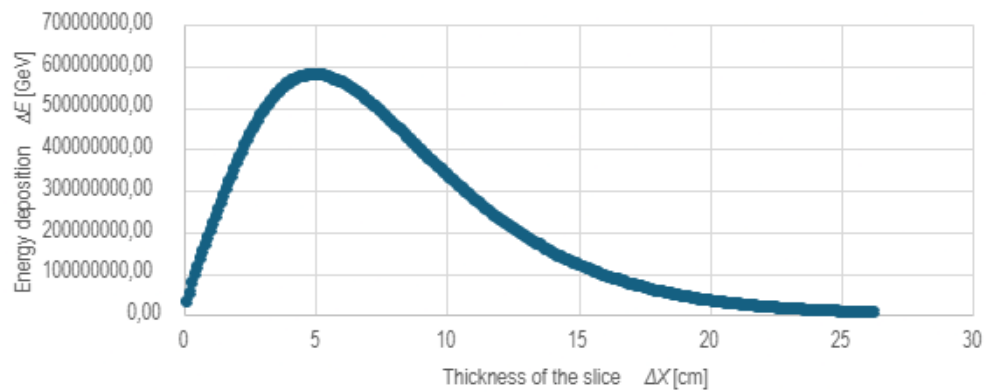
Longitudinal energy deposition profile for Pb target with 6 GeV energy of initial electron beam



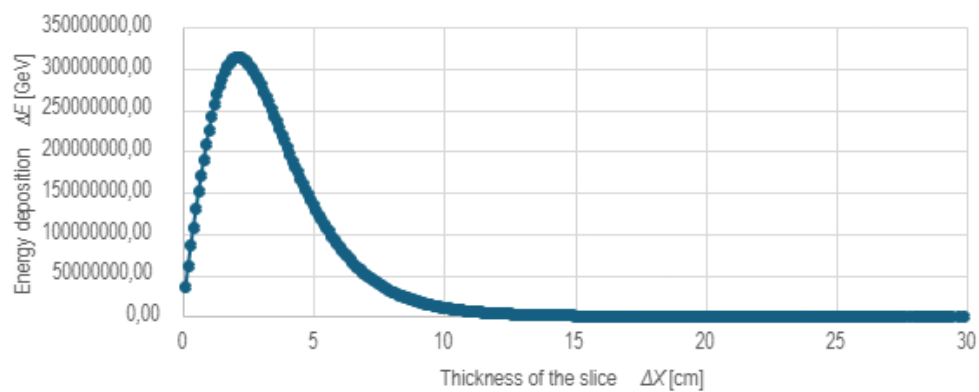
Cu:



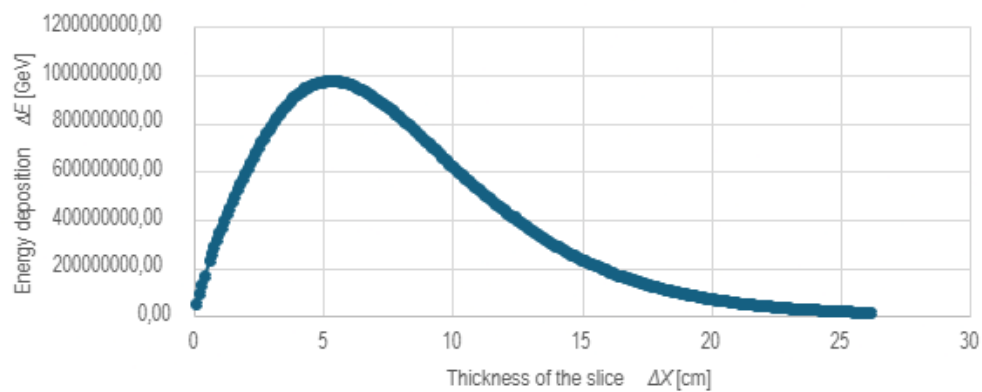
Longitudinal energy deposition profile for Cu target with 4 GeV
energy of initial electron beam



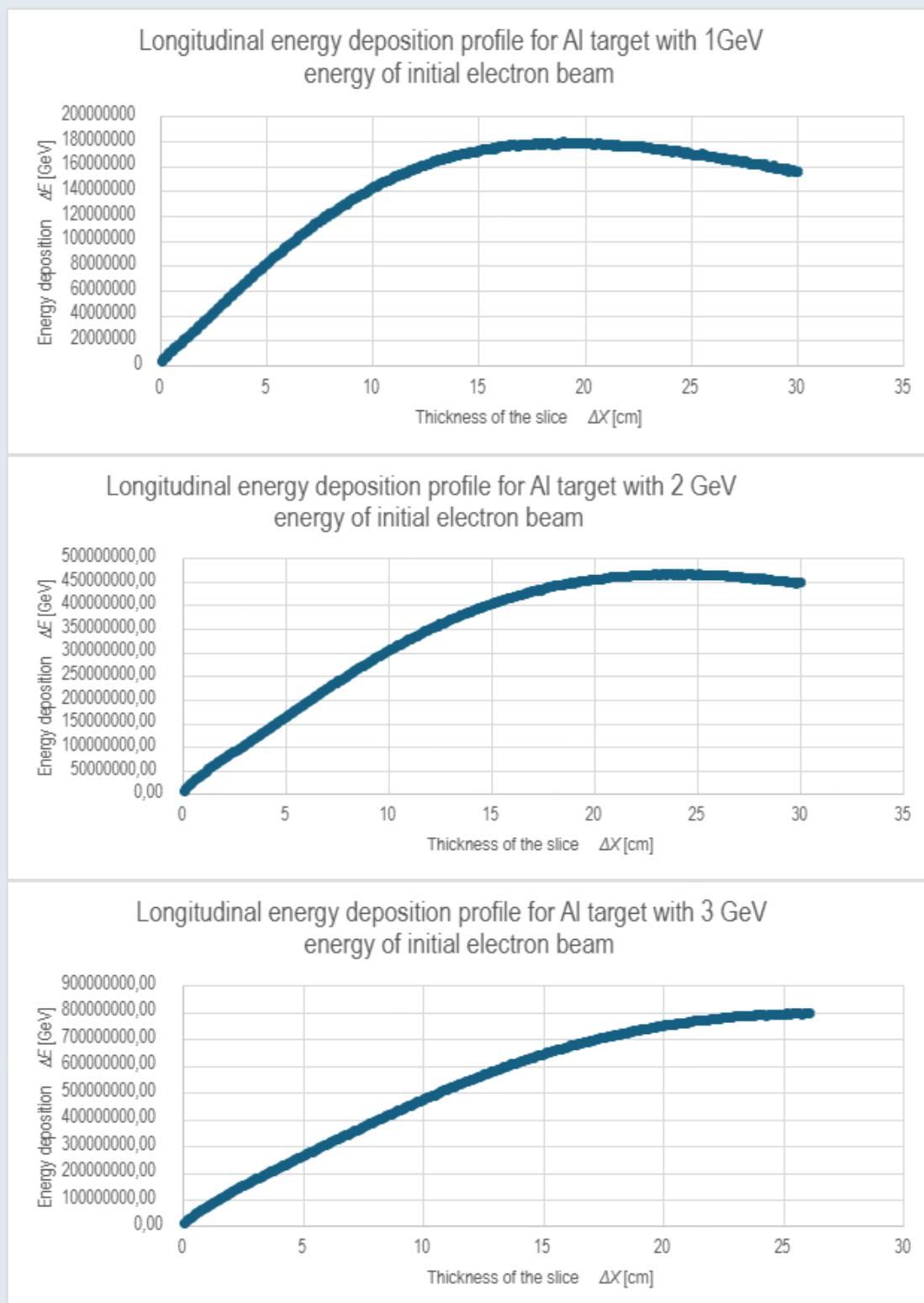
Longitudinal energy deposition profile for Cu target with 5 GeV
energy of initial electron beam



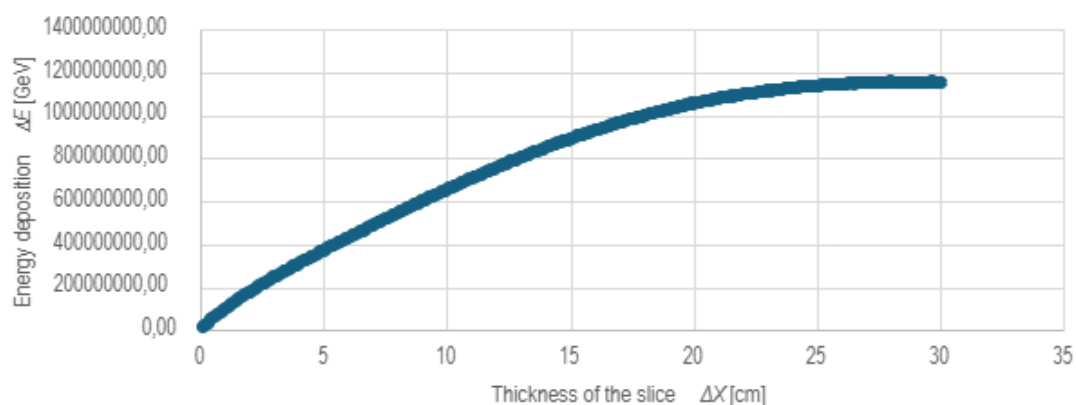
Longitudinal energy deposition profile for Cu target with 6 GeV
energy of initial electron beam



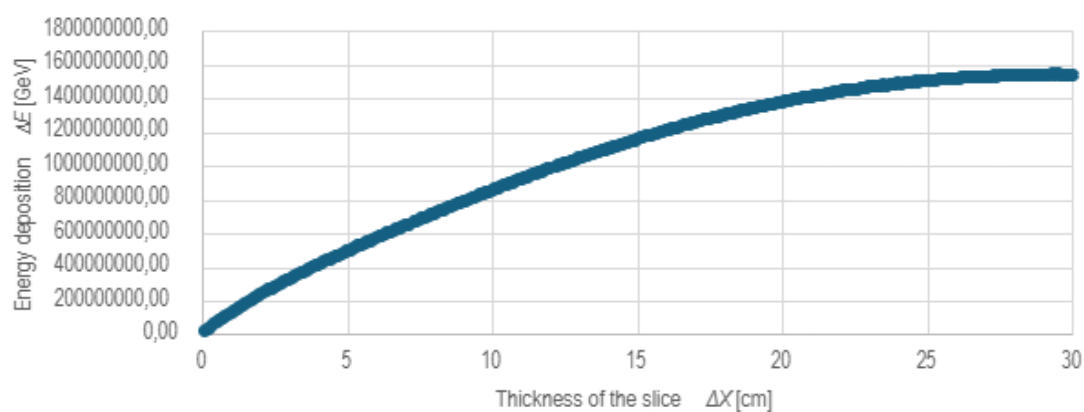
Al:



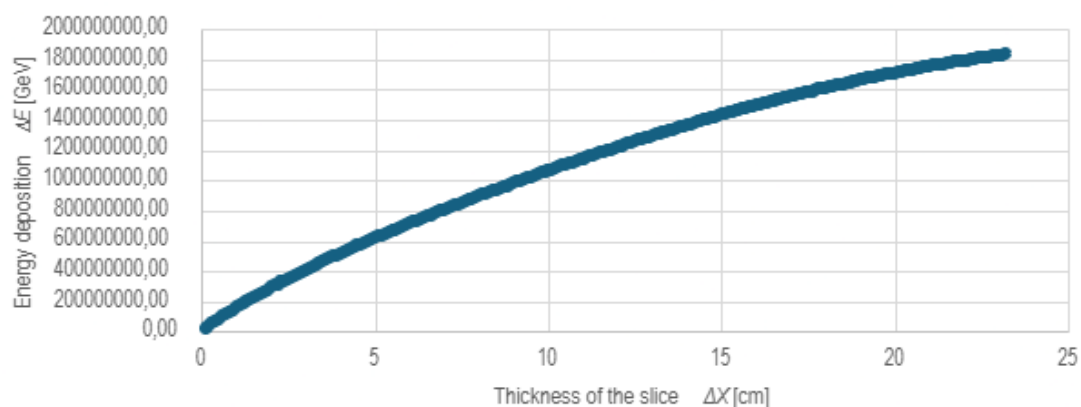
Longitudinal energy deposition profile for Al target with 4 GeV energy of initial electron beam



Longitudinal energy deposition profile for Al target with 5 GeV energy of initial electron beam



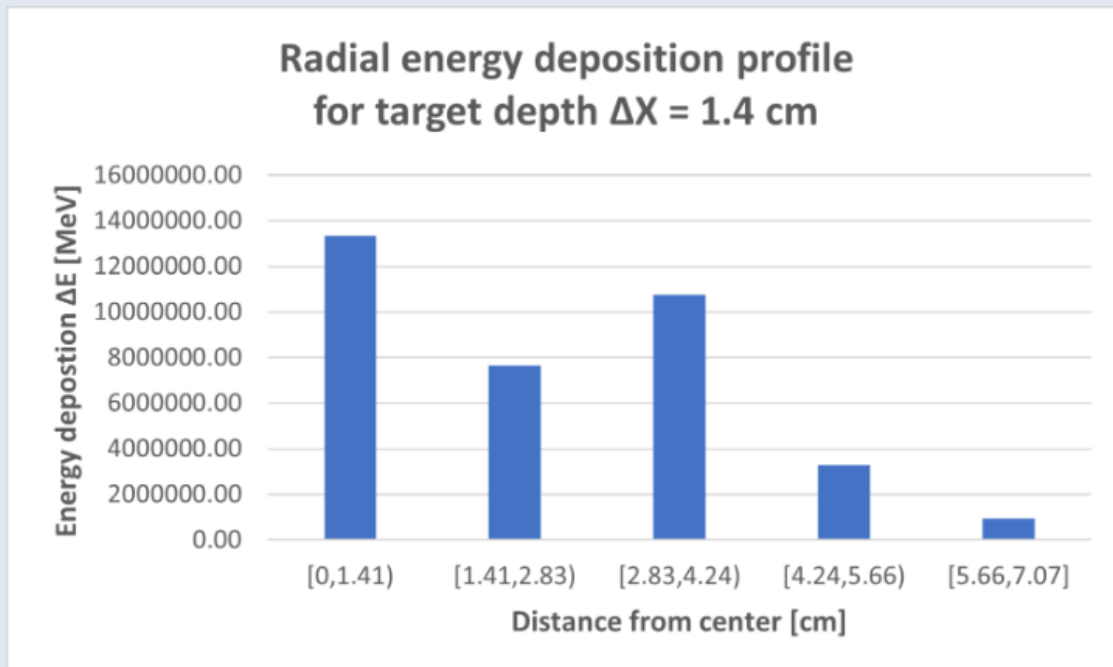
Longitudinal energy deposition profile for Al target with 6 GeV energy of initial electron beam



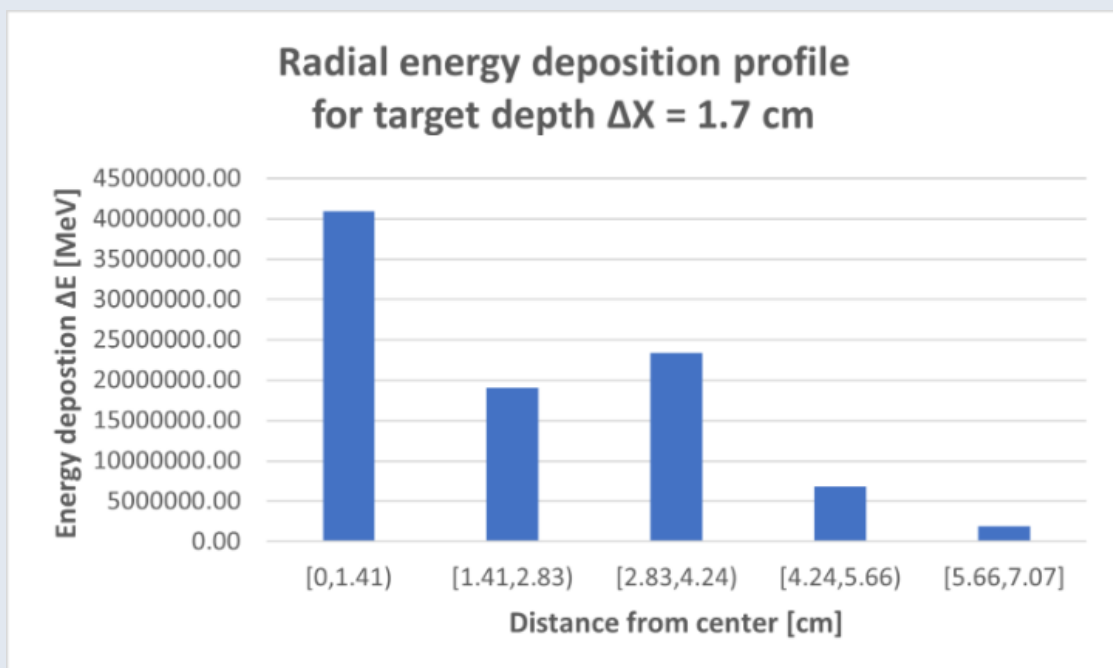
APPENDIX B – Radial Energy Deposition Profiles

Pb:

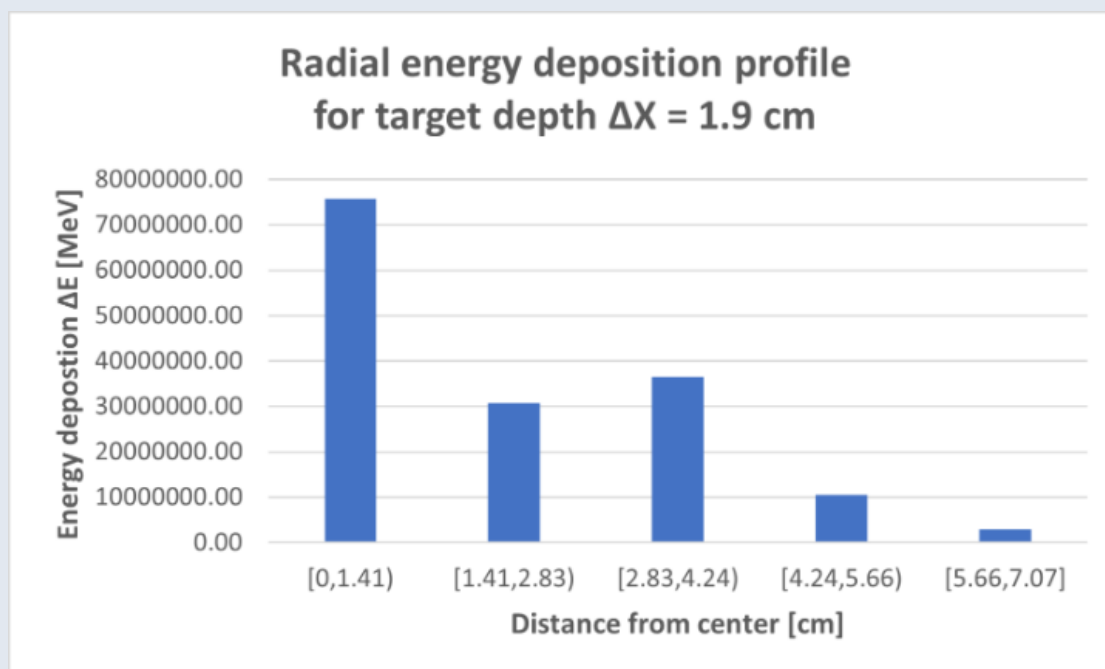
1 GeV:



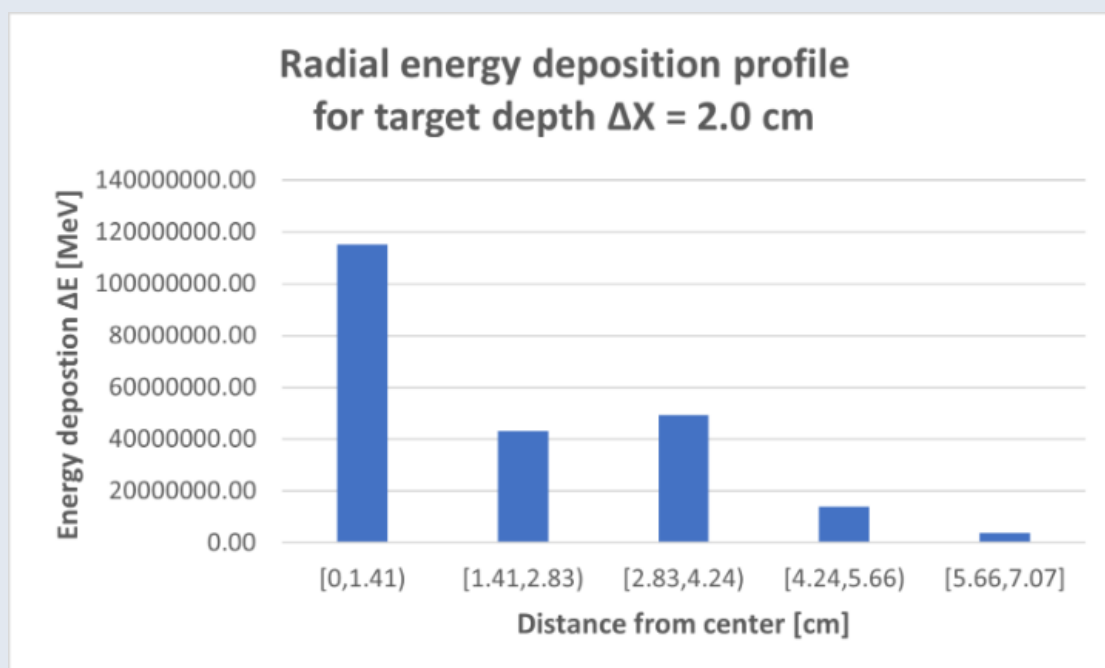
2 GeV:



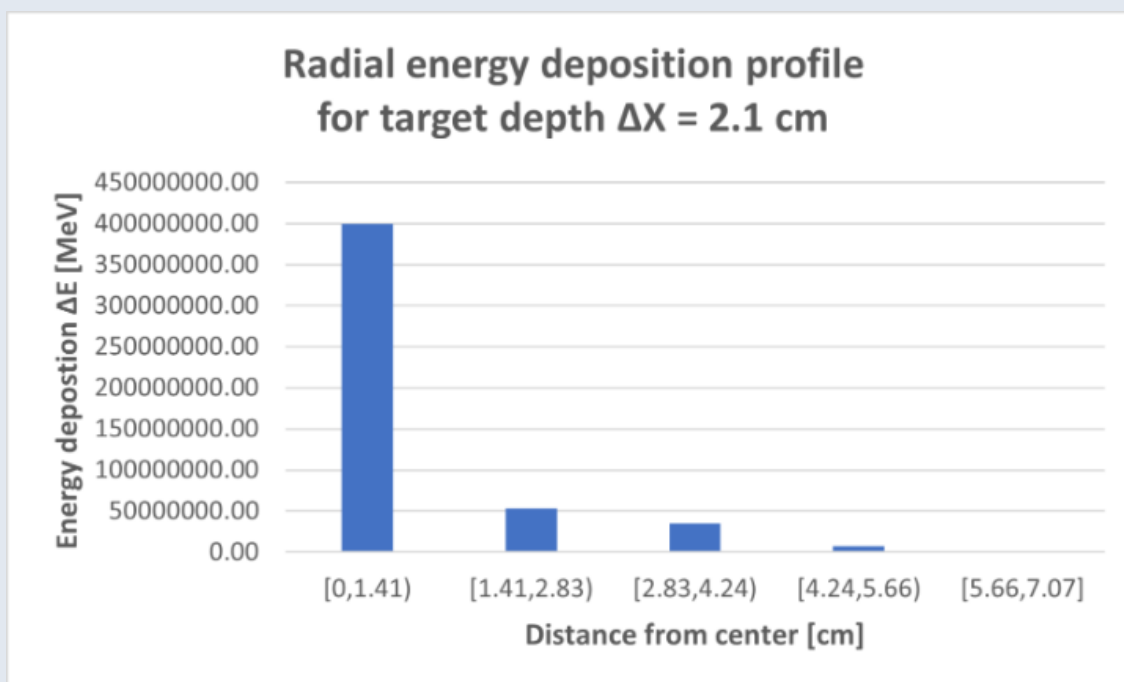
3 GeV:



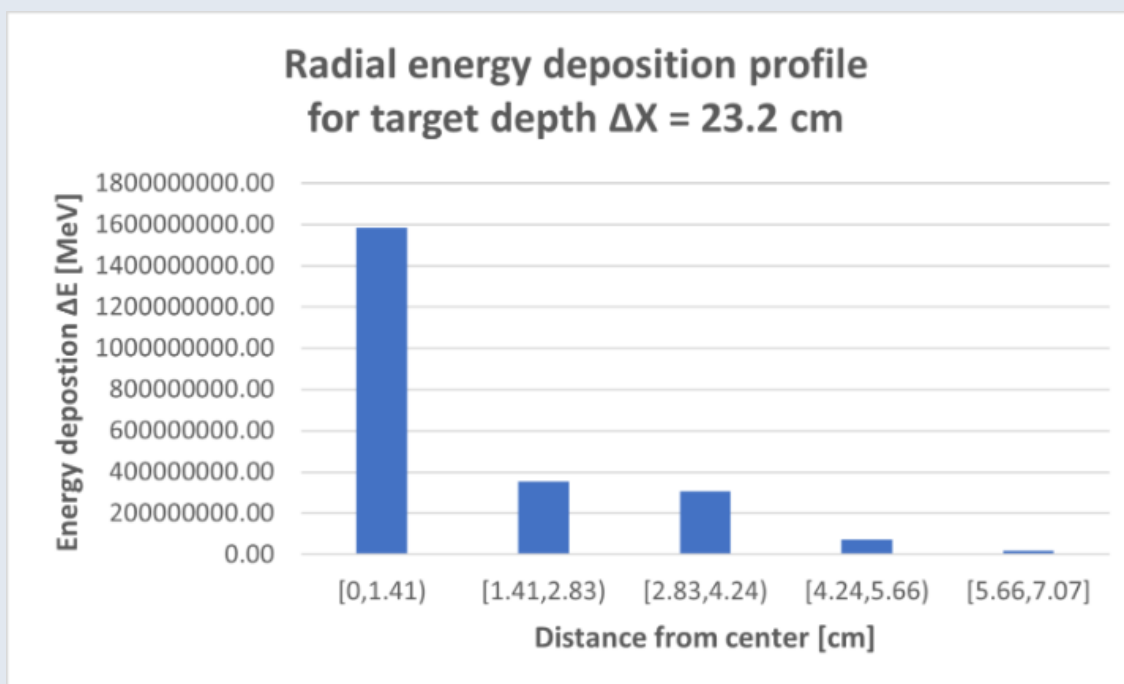
4 GeV:



5 GeV:

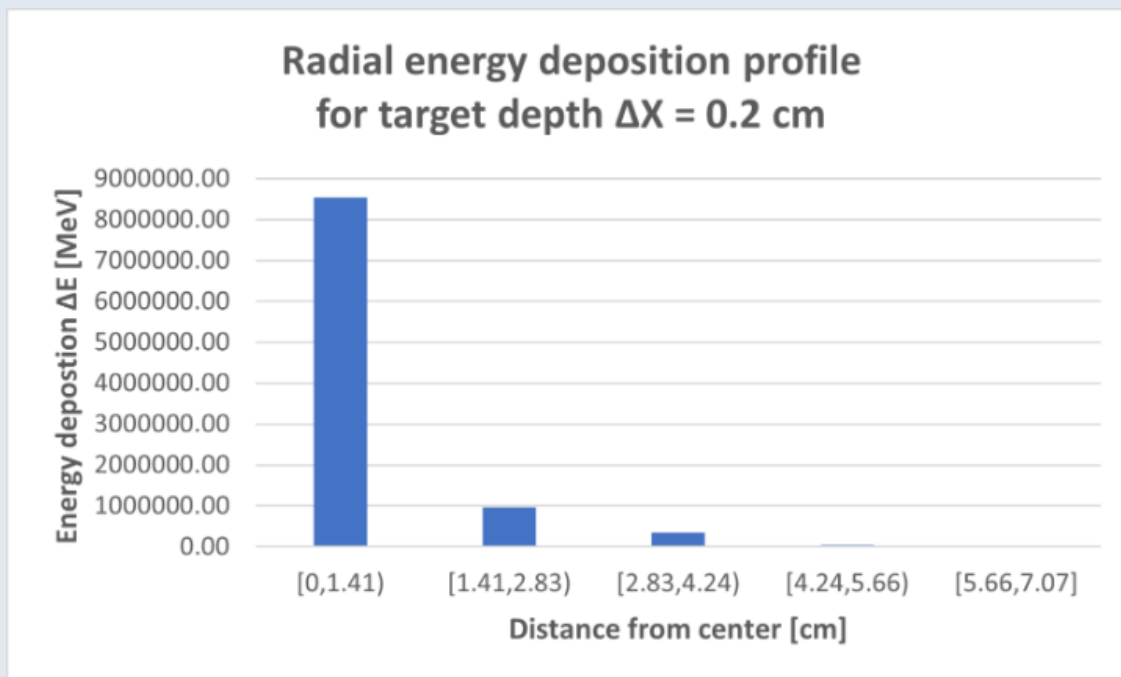


6 GeV:

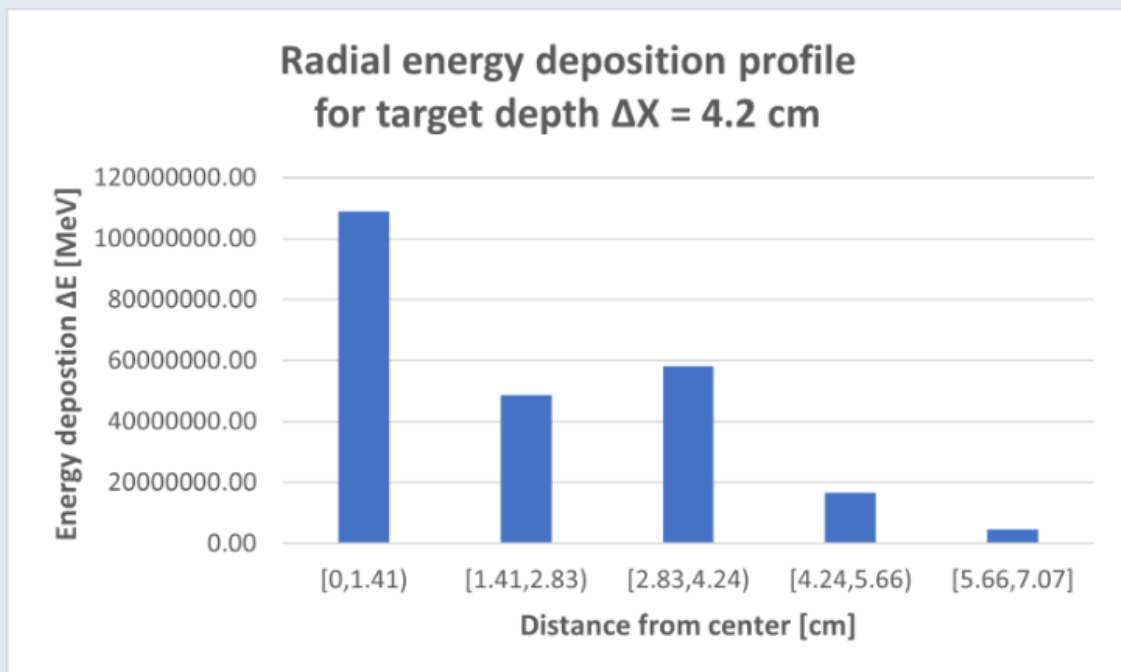


Cu:

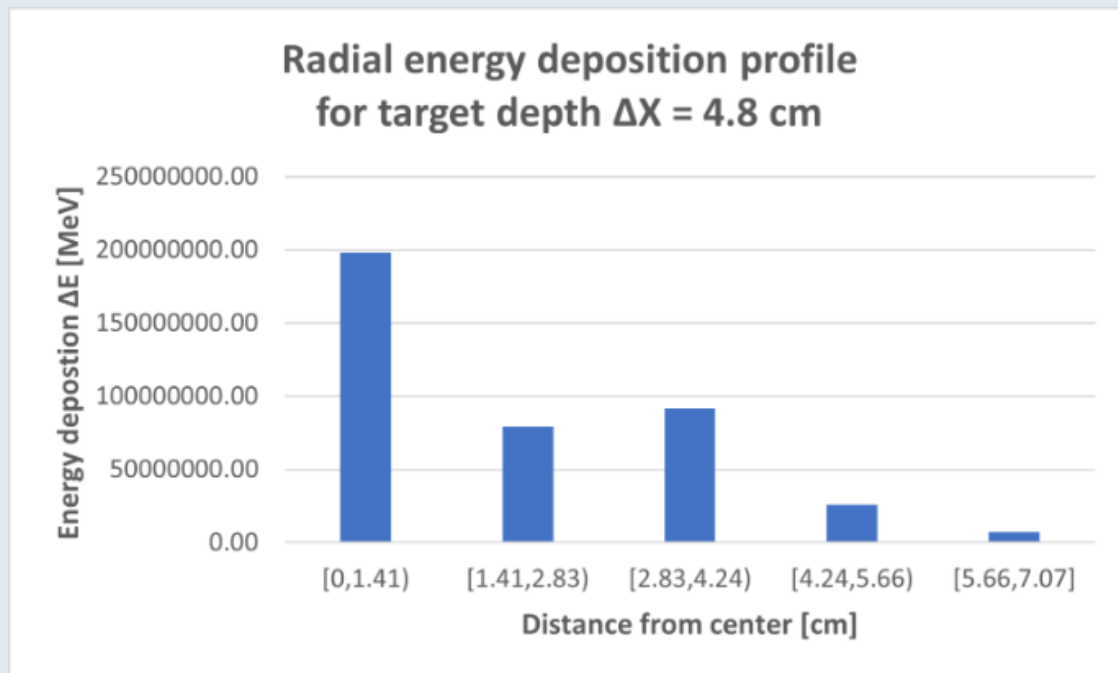
1 GeV:



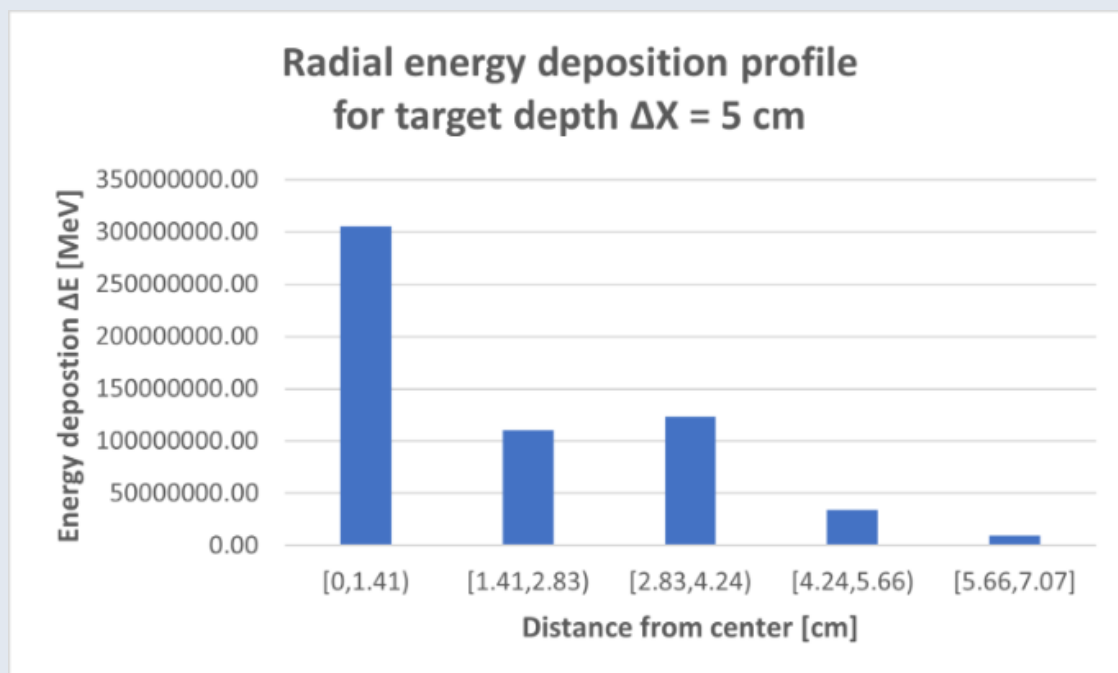
2 GeV:



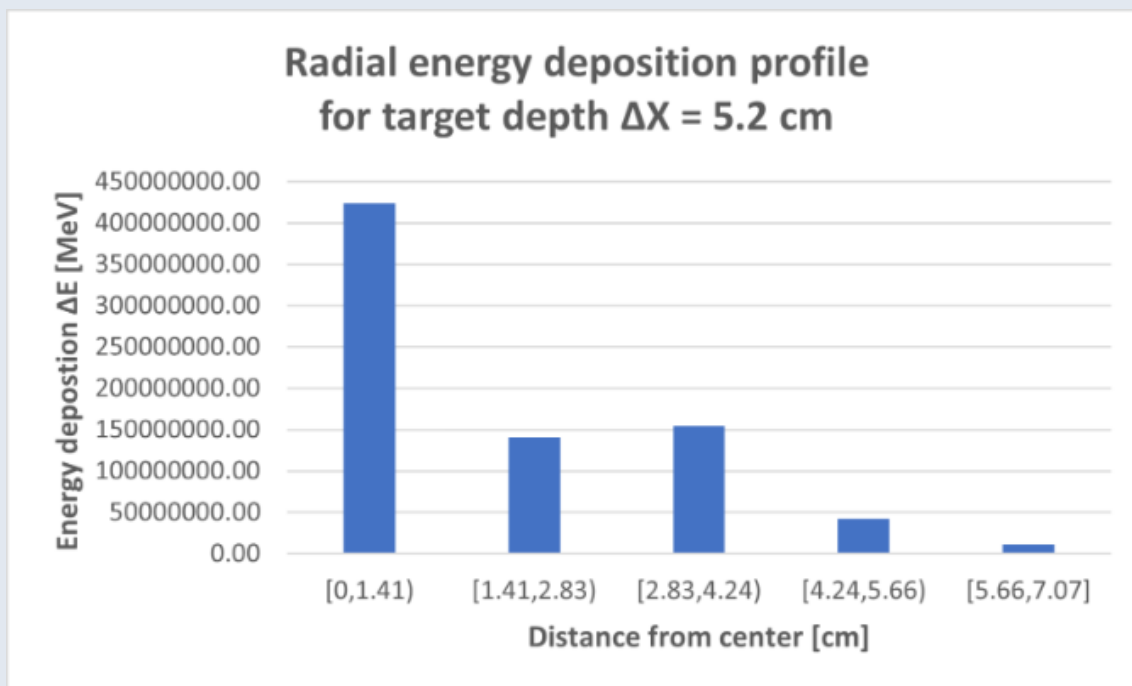
3 GeV:



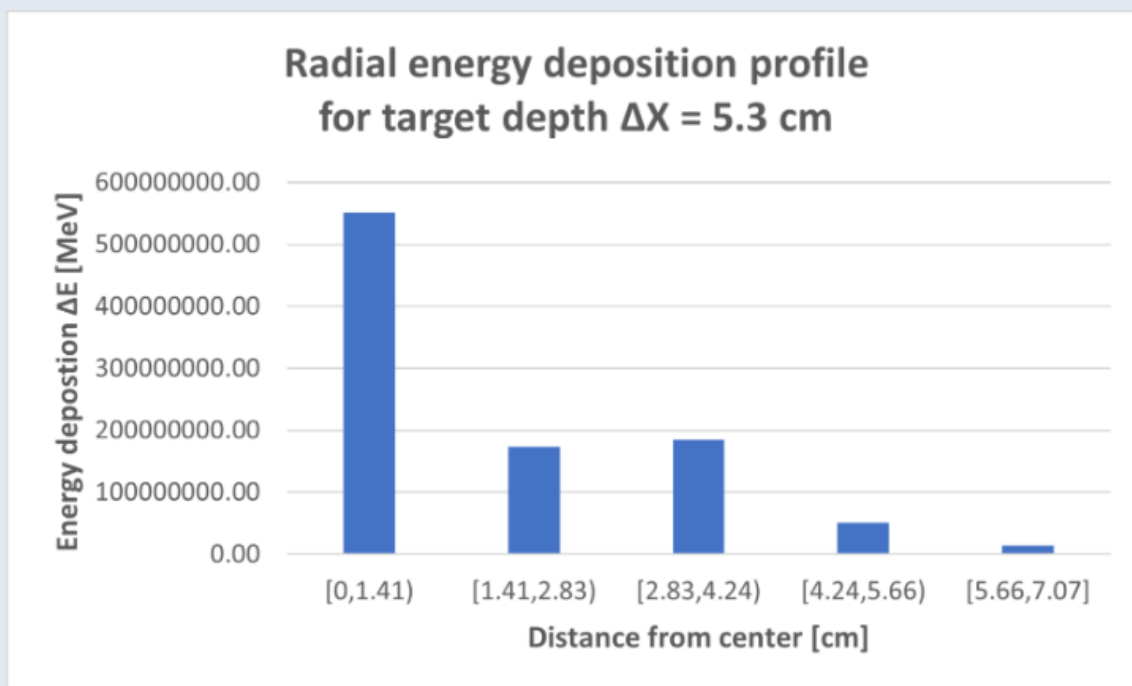
4 GeV:



5 GeV:

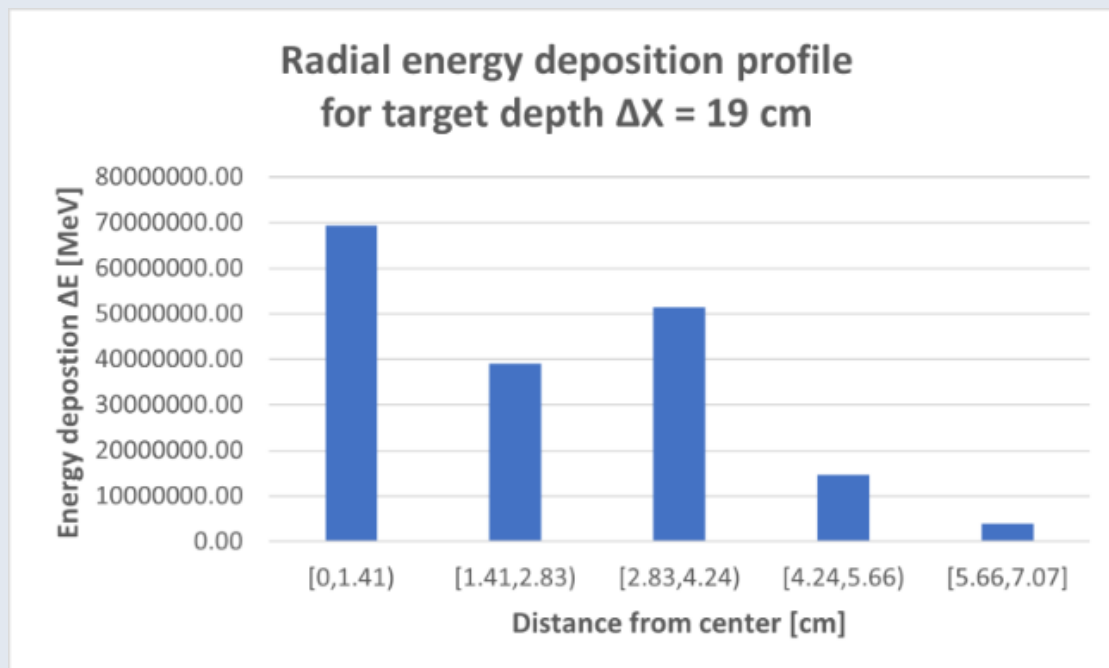


6 GeV:

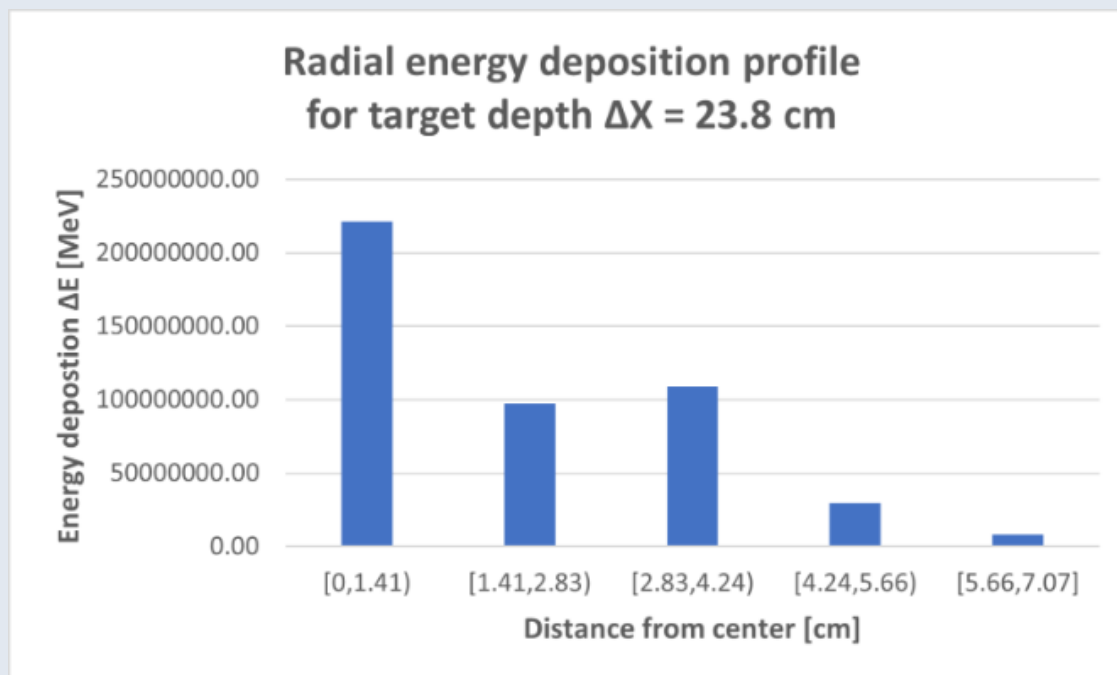


Al:

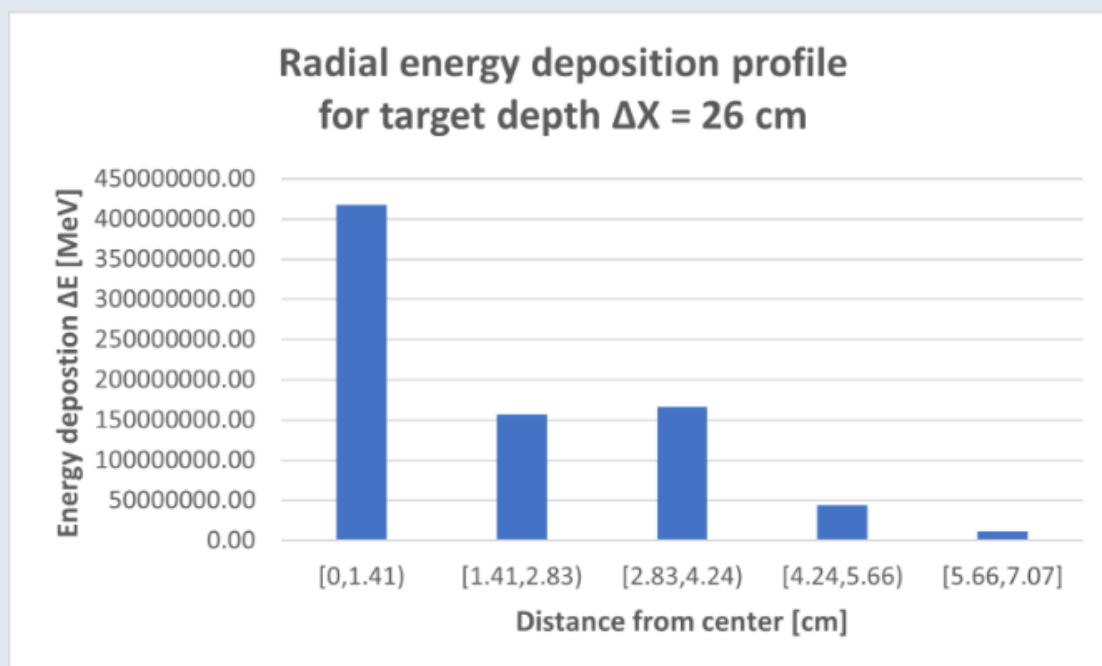
1 GeV:



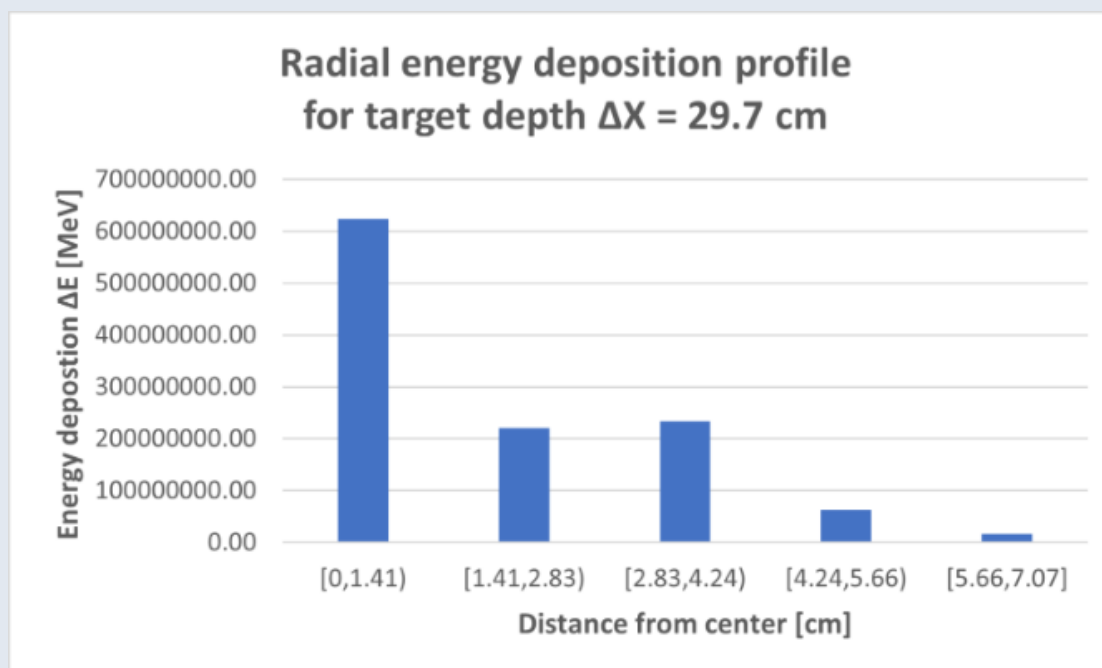
2 GeV:



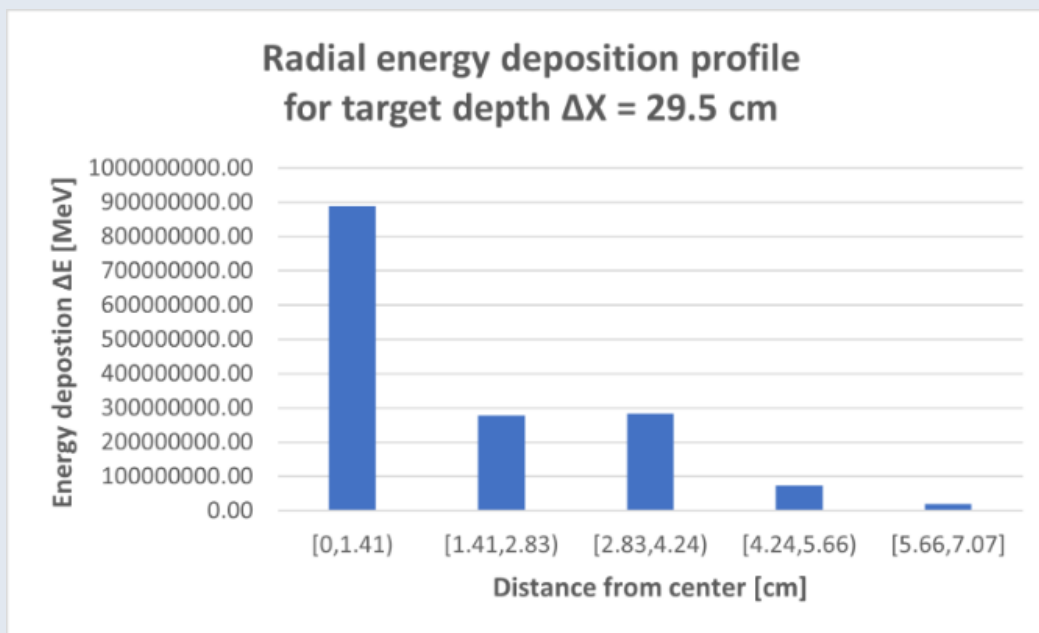
3 GeV:



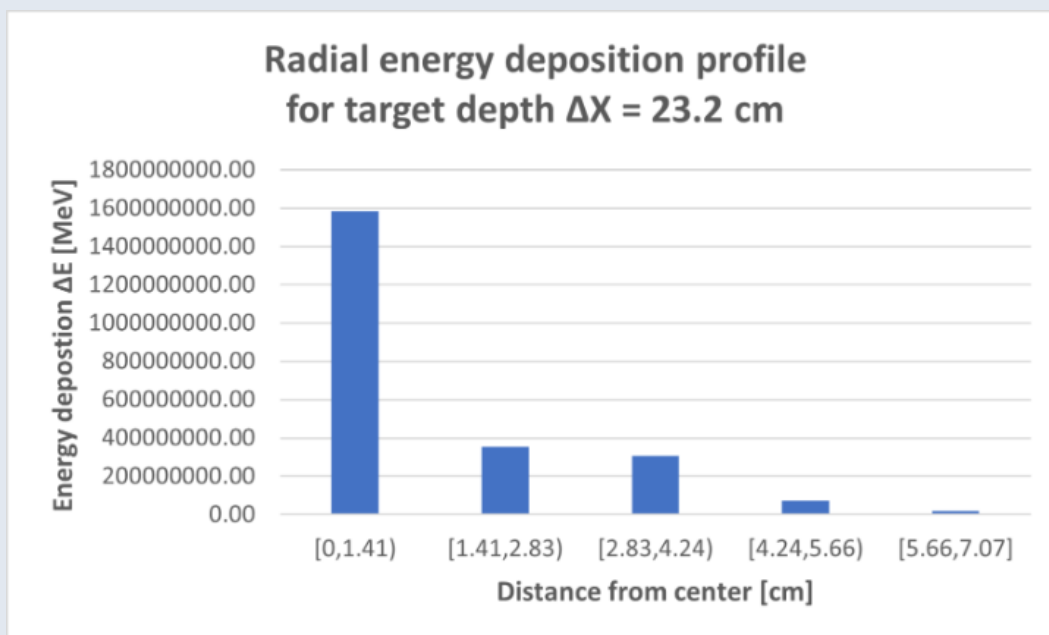
4 GeV:



5 GeV:

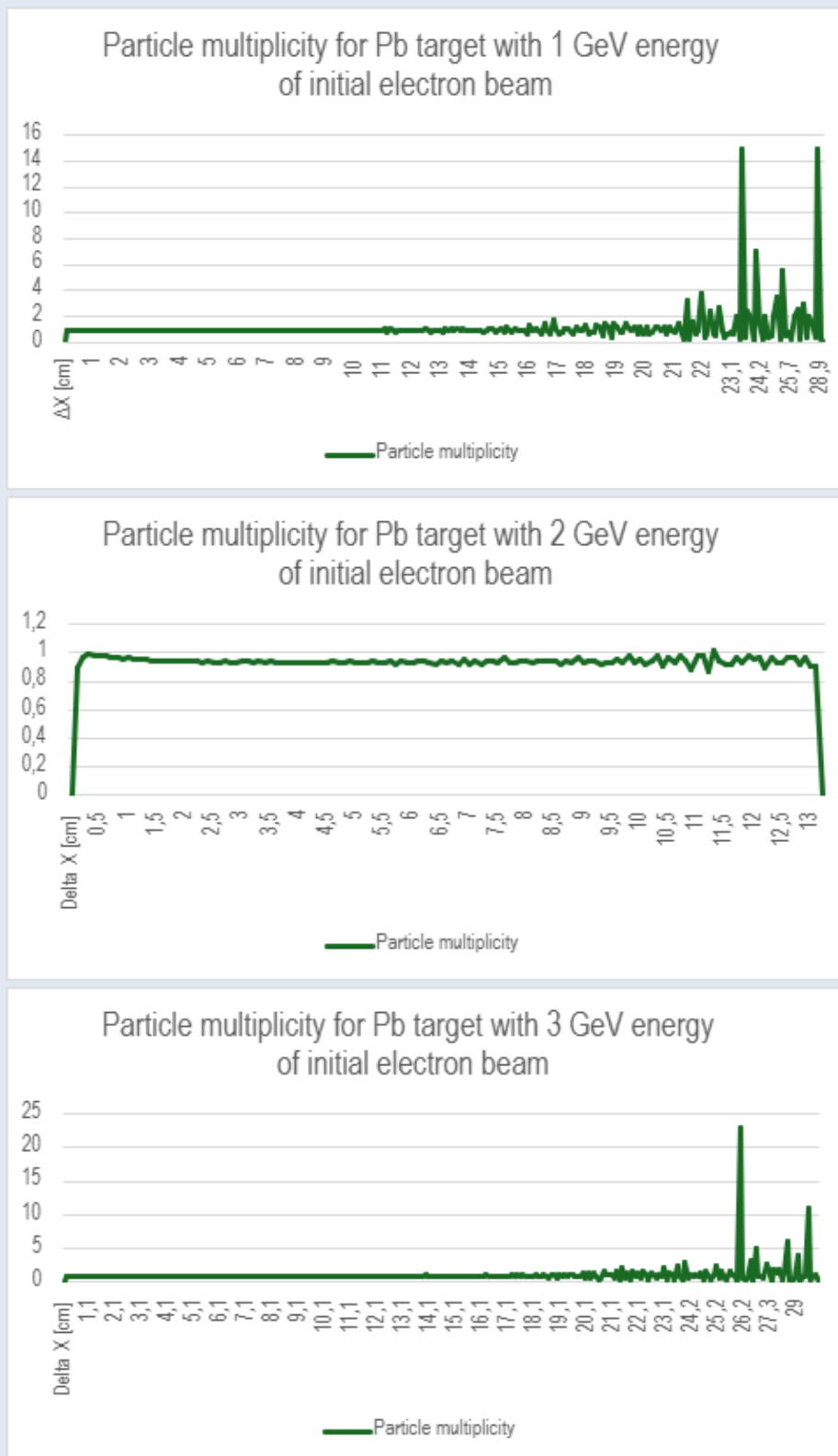


6 GeV:

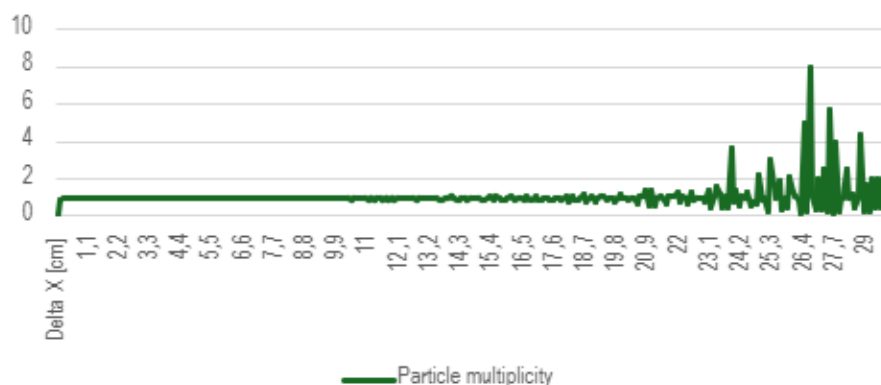


APPENDIX C – Multiplicity profiles

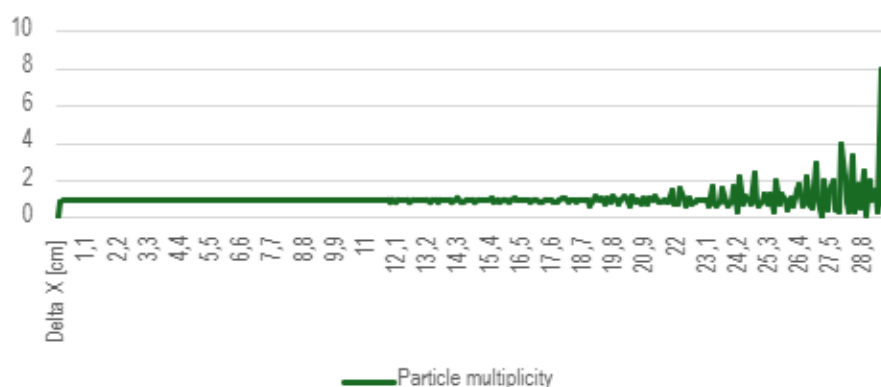
Appendix shows full multiplicity profiles, detailing particle distributions across events and enabling clear comparisons, highlighting patterns beyond the main results.



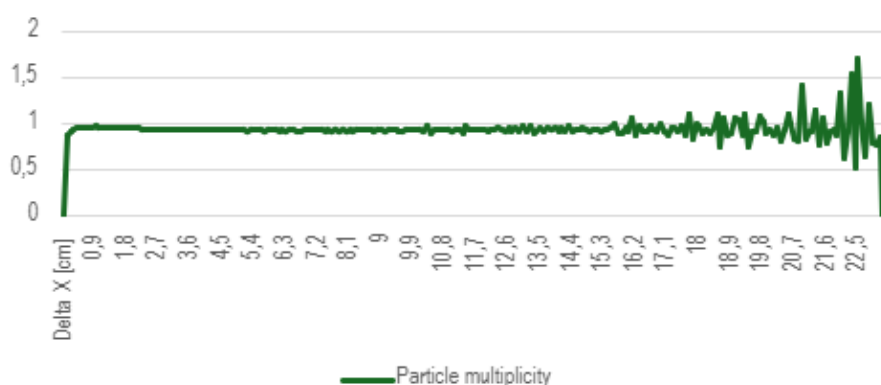
Particle multiplicity for Pb target with 4 GeV energy
of initial electron beam



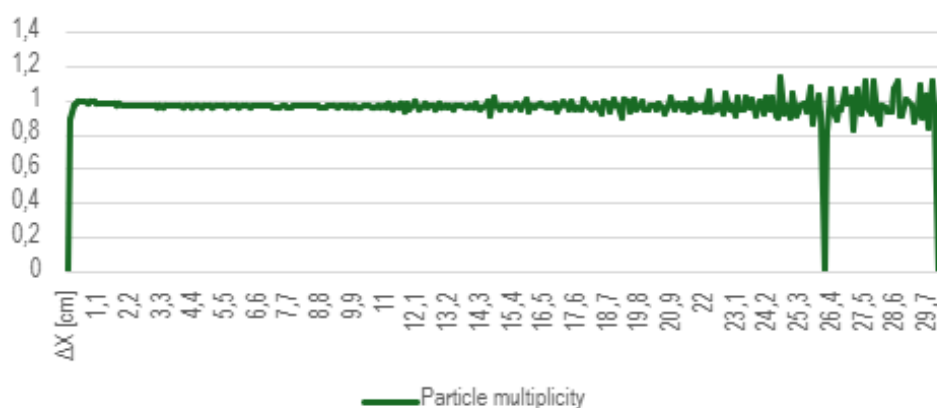
Particle multiplicity for Pb target with 5 GeV energy
of initial electron beam



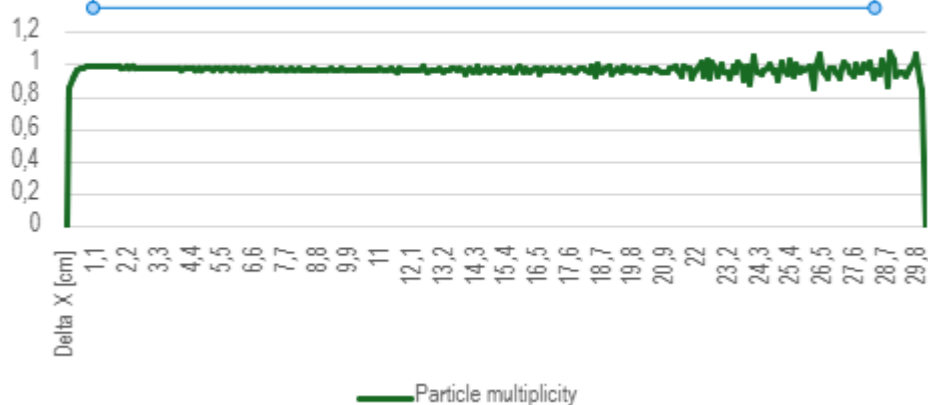
Particle multiplicity for Pb target with 6 GeV energy
of initial electron beam



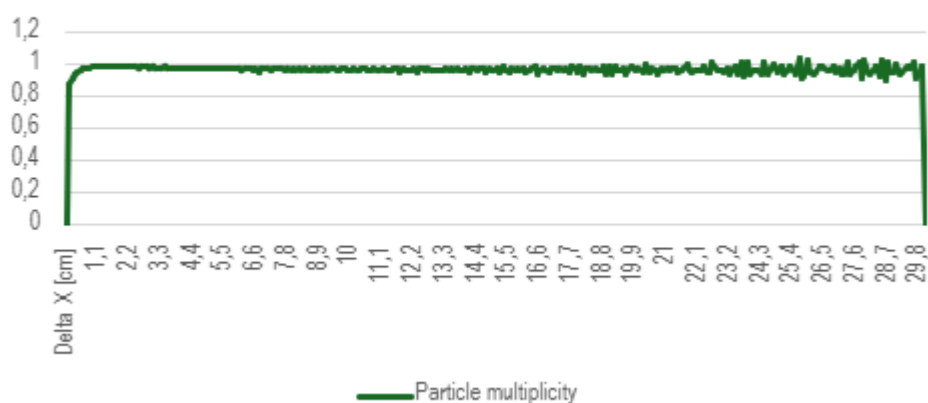
Particle multiplicity for Cu target with 1 GeV energy
of initial electron beam



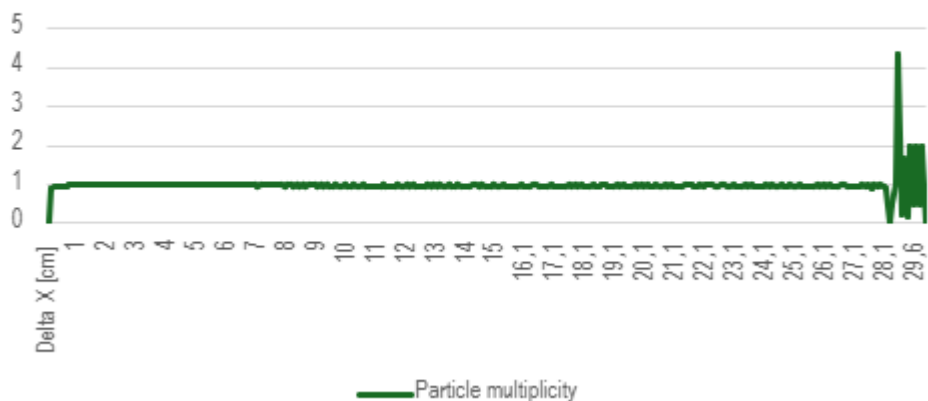
Particle multiplicity for Cu target with 2 GeV energy
of initial electron beam



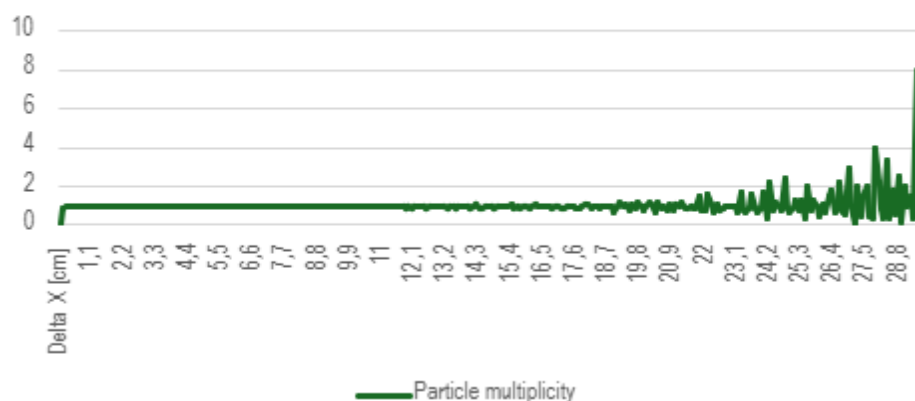
Particle multiplicity for Cu target with 3 GeV energy
of initial electron beam



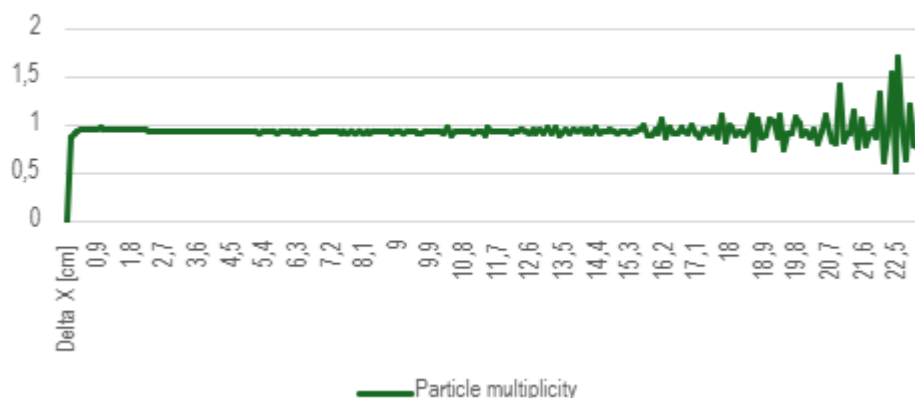
Particle multiplicity for Cu target with 4 GeV energy
of initial electron beam



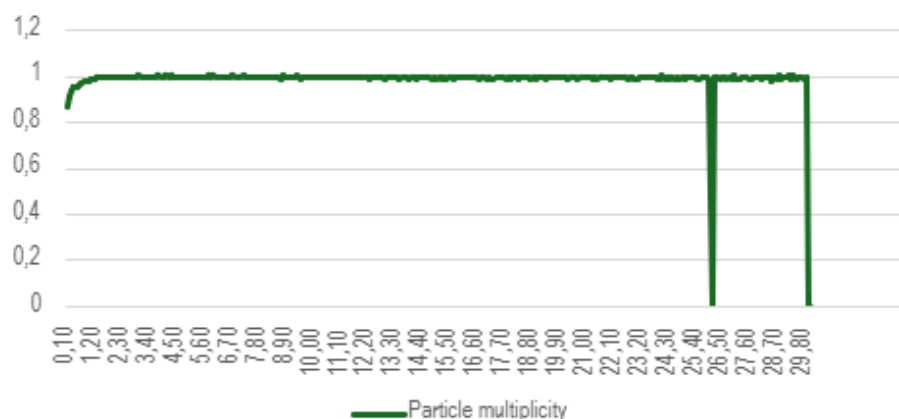
Particle multiplicity for Cu target with 5 GeV energy
of initial electron beam



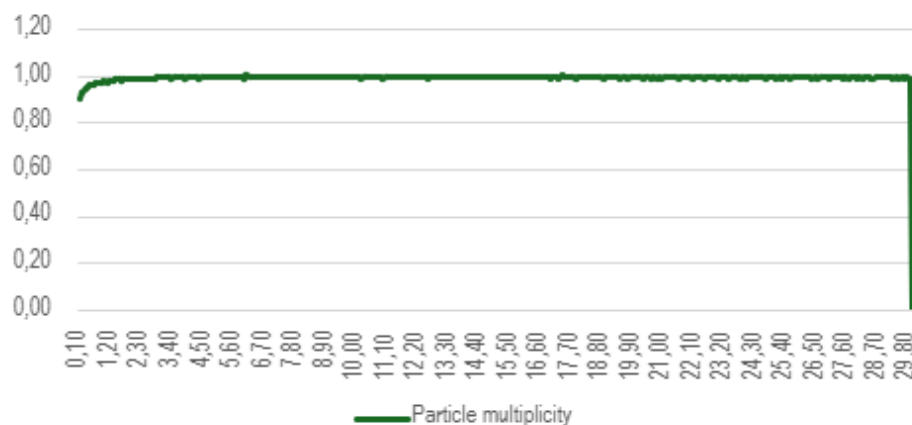
Particle multiplicity for Cu target with 6 GeV energy
of initial electron beam



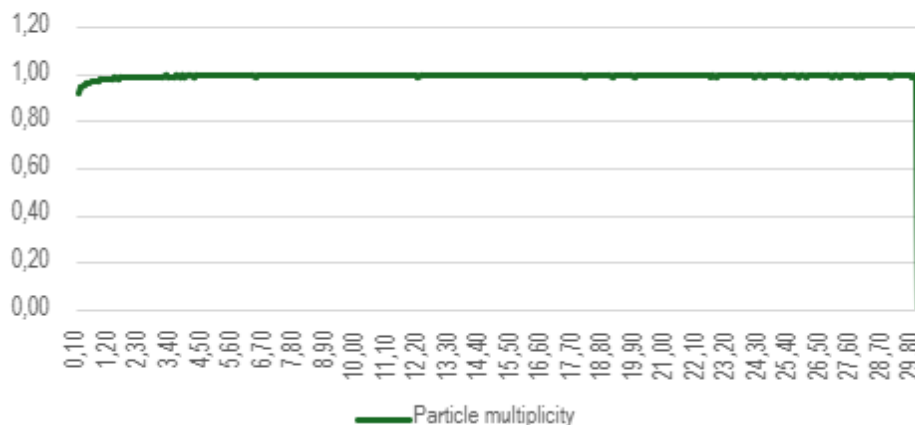
Particle multiplicity for Al target with 1 GeV energy
of initial electron beam



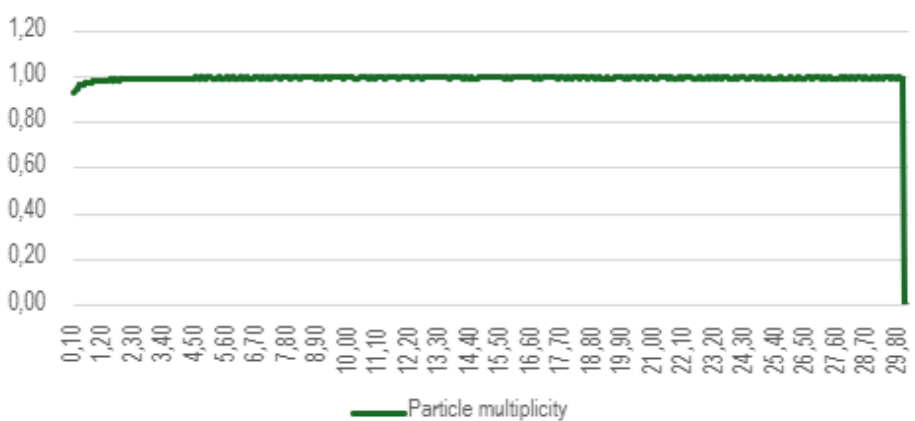
Particle multiplicity for Al target with 2 GeV energy
of initial electron beam



Particle multiplicity for Al target with 3 GeV energy
of initial electron beam



Particle multiplicity for Al target with 4 GeV energy
of initial electron beam



Particle multiplicity for Al target with 5 GeV energy
of initial electron beam

